



Investigating the protective properties of *Panax ginseng* and its constituents against biotoxins and metal toxicity: a mechanistic review

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Abstract

Natural toxins are toxic substances produced by living microorganisms and cause harmful effects to other creatures, but not the organisms themselves. Based on the sources, they are classified into fungal, microbial, herbal, algae, and animal biotoxins. Metals, the oldest toxicants, are not created or destroyed by human industry as elements, just concentrated in the biosphere. An antidote can counteract the toxic effects of a drug or toxin or mitigate the adverse effects of a harmful substance. The potential antidote effects of *Panax ginseng* in organ toxicity have been proved by many scientific research projects. Herein, we are going to gather a comprehensive mechanistic review of the antidotal effects of ginseng and its main constituents against natural toxins and metal toxicity. In this regard, a literate search has been done in PubMed/Medline, Science Direct, and Scopus from 2000 until 2024. The gathered data showed the protective impacts of this golden plant and its secondary metabolites against aflatoxin, deoxynivalenol, three-nitro propionic acid, ochratoxin A, lipopolysaccharide, nicotine, aconite, domoic acid, α -synuclein, amyloid β , and glutamate as well as aluminum, cadmium, chrome, copper, iron, and lead. These antidotal effects occur by multi-functional mechanisms. It may be attributed to antioxidant, anti-inflammatory, and anti-apoptotic effects. Future research directions on the antidotal effects of ginseng against natural toxins and metal toxicity involve broadening the scope of studies to include a wider range of toxins and metals, exploring synergistic interactions with other natural compounds, and conducting more human clinical trials to validate the efficacy and safety of ginseng-based treatments.

Keywords Antidote · Cell signaling · Ginsenoside · Mechanism · Metal · Toxicants

Abbreviations

AP-1	Activator protein 1
A β	Amyloid β
AST	Aspartate aminotransferase
ATF4	Activating transcription factor 4
ATFB-1	Aflatoxin-B1
ATF4	Activating transcription factor 4
Akt	Protein kinase B
BACE1	β -Amyloid cleavage enzyme 1
ALT	Alanine transaminase
Bcl-2	B-cell lymphoma 2
Al	Aluminum
Bax	Bcl-2-associated X protein
BDNF	Brain-derived neurotrophic factor
cAMP	Cyclic adenosine monophosphate
CAT	Catalase
Cd	Cadmium
CDK5	Cyclin-dependent kinase 5
CHOP	CCAAT/enhancer-binding protein homologous protein
COXs	Cyclooxygenases
Cr	Chromium

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CREB	CAMP response element-binding protein
eIF2 α	Eukaryotic translation initiation factor 2
ERK	Extracellular signal-regulated kinase
FGFs	Fibroblast growth factor
GABA	Gamma-aminobutyric acid
GPx	Glutathione peroxidase
GSH	Glutathione
HO-1	Heme oxygenase-1
IFN-	Interferon-
AIM2-associated inflammasome	Absent in melanoma 2-associated inflammasome
IL-	Interleukin-
iNOS	Inducible nitric oxide synthase
JNKs	Jun NH2-terminal kinases
Keap1	Kelch-like ECH-associated protein 1
LDH	Lactate dehydrogenase
LDL	Low-density lipoprotein
LPS	Lipopolysaccharide
LPAR	Lysophosphatidic acid receptor
MAPK	Mitogen-activated protein kinase
MDA	Malondialdehyde
NF- κ B	Nuclear factor κ -light-chain enhancer of activated B cells
NLRP3	Nucleotide oligomerization domain (NOD)-like receptor protein-3
Nrf2	Nuclear factor erythroid 2-related factor 2
Pb	Lead
PERK	PKR-like endoplasmic reticulum kinase
p-GSK3	Phosphorylated glycogen synthase kinase-3
PI3K	Phosphatidylinositol 3-kinase
PKA	Protein kinase A
PP2A	Protein phosphatases 2A
PPAR	Peroxisome proliferator-activated receptor
ROS	Reactive oxygen species
Sirt1	Sirtuin 1
SOD	Superoxide dismutase
TGF- β 1	Transforming growth factor- β 1
TLR	Toll-like receptor

TNF- α
3-NP

Tumor necrosis factor- α
3-Nitropropionic acid

Introduction

Herbal medicines have been commonly used in many traditional medical systems to treat numerous disorders (Ang-Lee et al. 2001). One of the most widely used medicinal herbs is ginseng, which refers to the dried rhizomes of *Panax ginseng* Meyer (Korean ginseng), *P. notoginseng* (Burk.) FH Chen (Notoginseng), and *P. quinquefolium* L. (American ginseng) (Kim 2012). *P. ginseng*, a perennial herbaceous plant from the Araliaceae family, is the most extensively used ginseng mainly cultivated in China, Japan, Korea, and Russia (Lü et al. 2009). With more than 2000 years of history, this herb has been popular in traditional medicine systems, such as traditional Chinese medicine and Ayurveda, due to its remarkable health-promoting properties (Kim 2012). The earliest pharmacological description of ginseng was introduced by Shennong Bencao Jing in the first century. It described the nourishing and tonifying effects of ginseng on the spleen, lungs, heart, kidney, and liver. Additionally, it mentioned that ginseng could normalize pulse rate and improve sedation. Other pharmacological aspects of ginseng that were discussed in this old herbal textbook include the improvement of visual and mental activity, the enhancement of longevity, and the decline of pathological factors. Another herbal medicine textbook, Bencaojing Jizhu, explained the therapeutic effects of ginseng on cognitive function, blood circulation, internal coldness, the sensation of thirst and chest fullness, pain in the chest and abdomen, vomiting, and diarrhea. The pre-modern herbal text Bencao Gangmu also explained these pharmacological aspects of ginseng. These examples of traditional herbal texts describe the pharmacological effects of ginseng in curing weakness, vomiting, diarrhea, anxiety, and mental well-being. All of these traditional prescriptions demonstrate the protective effects of ginseng on vital organs (Park et al. 2012). Modern pharmacological experiments have proven the therapeutic effects of ginseng in a wide range of health conditions. Ginseng is used to improve the immune system, increase energy levels, and enhance overall well-being (Ratan et al. 2021). Research has demonstrated its potential in managing and protecting against cardiovascular diseases (Yu et al. 2022), diabetes (Aminifard et al. 2021), cancer (Zhao et al. 2022), immune deficiency (He et al. 2018), sexual dysfunctions (Ghorbani et al. 2019), and neurodegenerative disorders (Zanuso et al. 2022). Its benefits in neurodegenerative diseases have resulted in many patent reports (Rajabian et al. 2019). Additionally, ginseng has been studied for its effects on mental health, including its potential as an antidepressant and anxiolytic agent (Lee and Rhee 2017).

From a mechanistic viewpoint, ginseng induces modulatory effects on toll-like receptors (TLRs), cyclooxygenases (COXs), nitric oxide synthases (NOSs), reactive oxygen and nitrogen species (RONS), and antioxidant enzymes, including catalase (CAT), superoxide dismutase (SOD), and glutathione peroxidase (GPx) (Yoon et al. 2021). Furthermore, it has been reported as a robust anti-inflammatory agent through the suppression of critical inflammatory processes, including nuclear factor-kappa B (NF- κ B) and mitogen-activated protein kinase (MAPK) (Li et al. 2020). It has hepatoprotective effects and regulates the levels of hepatic marker enzymes, including lactate dehydrogenase (LDH), aspartate aminotransferase (AST), and alanine aminotransferase (ALT) (Simental-Mendía et al. 2021). Also, it has shown a good impact on sperm maturation and testicular functions by reducing malondialdehyde (MDA) and increasing the levels of enzymatic/nonenzymatic antioxidants (Ramesh et al. 2012).

Ginseng contains various bioactive compounds such as phytosterols, sesquiterpenes, flavonoids, polyacetylenes, alkaloids, phenolic compounds, and ginsenosides (Park et al. 2018). Among these, ginsenosides, which are a type of saponins, have been extensively studied for their anti-inflammatory (Yi 2021), anti-cancer (Jung et al. 2023), and neuroprotective properties (Qi et al. 2022). These compounds interact

with molecular targets in the body, modulate signaling pathways, and have therapeutic effects (Lü et al. 2009). Ginsenosides are categorized into three main groups based on their structural variances: protopanaxadiols (Rb1, Rb2, Rc, Rd, Rg3, Rh2, and Rh3), protopanaxatriols (Re, Rf, Rg1, Rg2, and Rh1), and pentacyclic ginsenosides, including oleanolic acid-type Ro ginsenoside (Park et al. 2018) (Figs. 1 and 2).

Toxicants are chemicals produced naturally or synthesized by human activities. Naturally occurring chemicals are named toxins which are metabolic products of fungi, microbes, plants, algae, and animals to protect them against pathogens or predators. It should be considered that a toxicant could not be a toxin, but a toxin is a toxicant (Cope and Hodgson 2010). Metals are naturally occurring substances on Earth. Anthropological activities increase their concentration on the site of human living. The inhibition or activation of special enzymes, disruption of the structure, and function of subcellular organelles, carcinogenicity, and multi-organ damage are the ways metals induce toxicity (Cope and Hodgson 2010).

An antidote is a substance that counteracts or mitigates the toxic effects of a drug or toxin (Chacko and Peter 2019). Numerous scientific research projects have demonstrated the antidotal and protective effects of ginseng against organ toxicity (Wang et al. 2022). Ginseng may enhance the body's

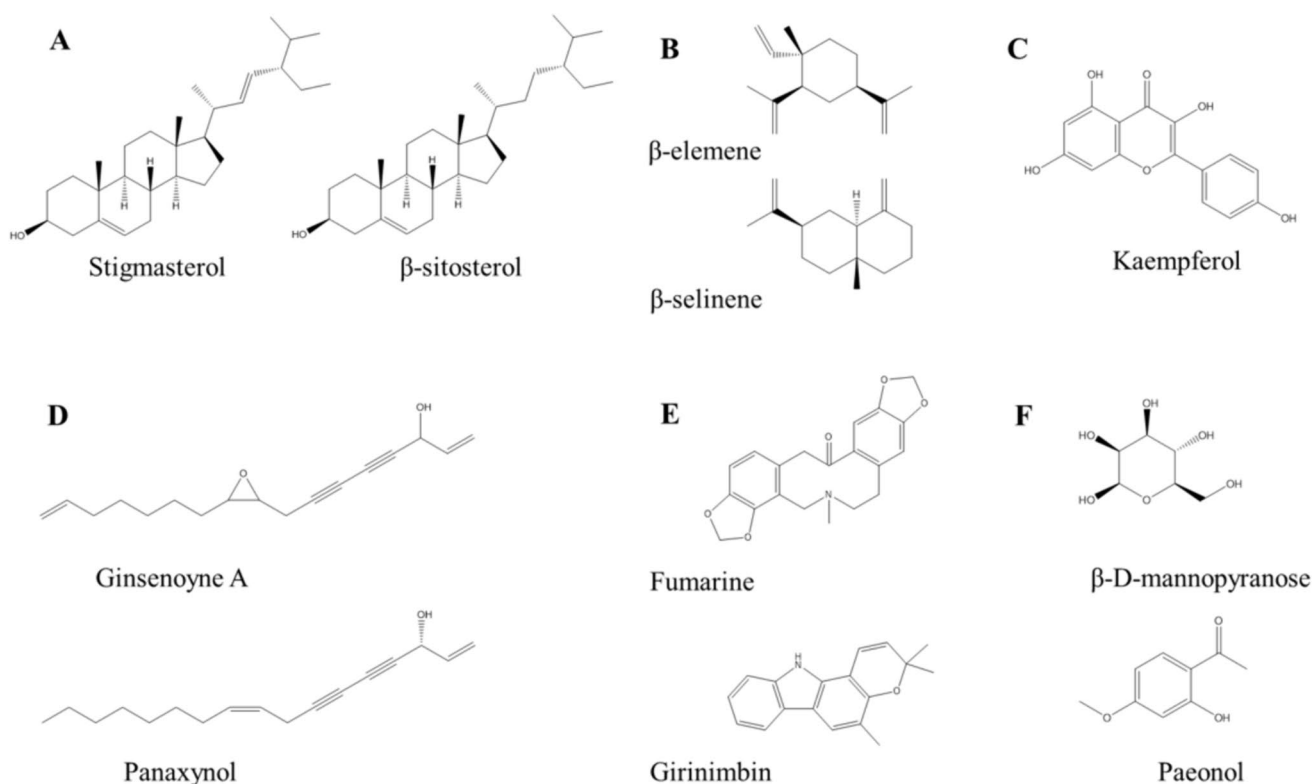


Fig. 1 Chemical structure of selected non-saponin active constituents detected in *P. ginseng* including **A** phytosterol, **B** sesquiterpenes, **C** flavonoids, **D** polyacetylenes, **E** alkaloids, **F** phenolic compounds

Fig. 2 Chemical structure of selected saponin active constituents detected in *P. ginseng* including protopanaxadiols (**A**), protopanaxatriols (**B**), and oleanane-type ginsenosides (**C**)

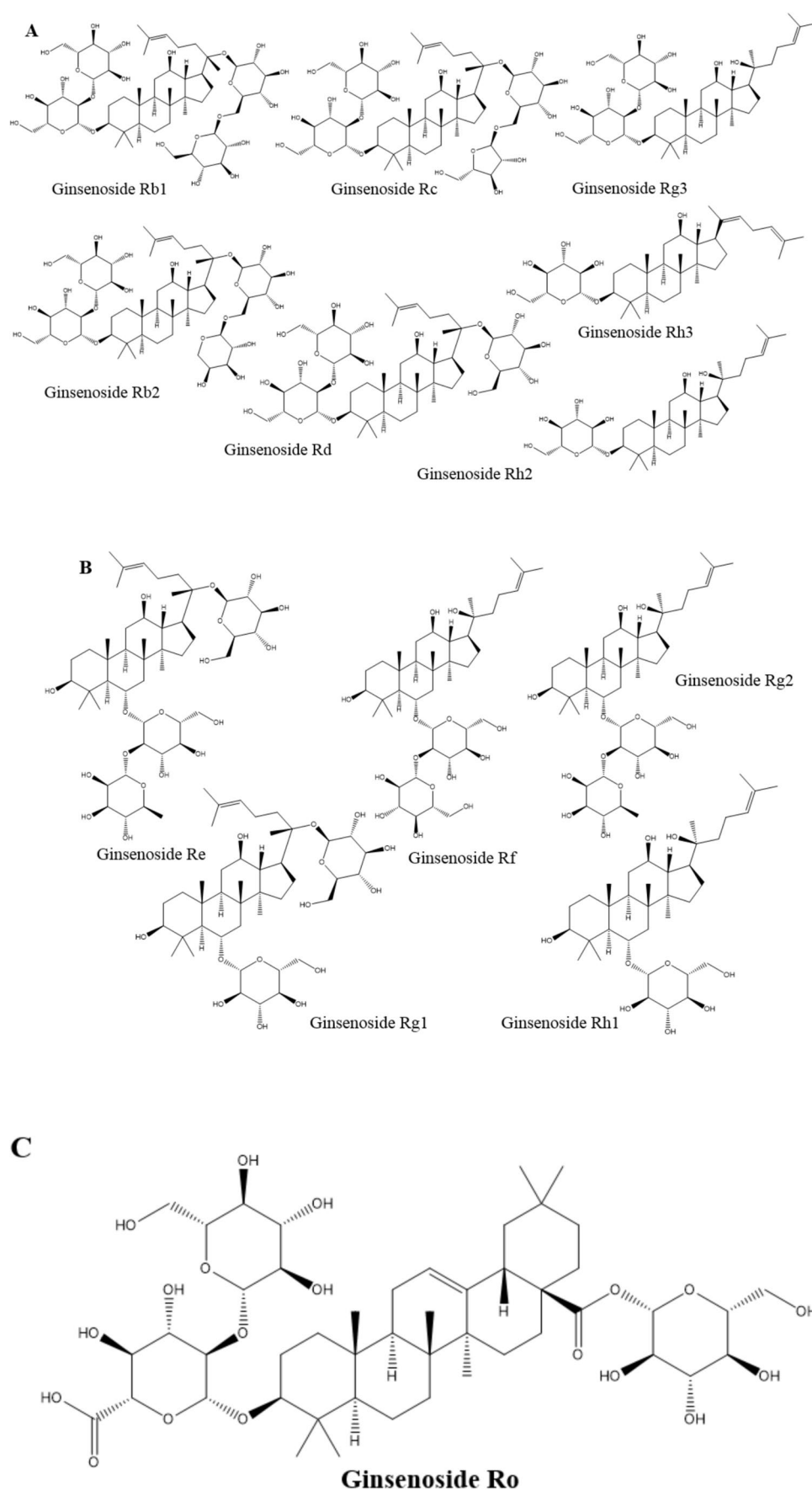


Fig. 2 (continued)

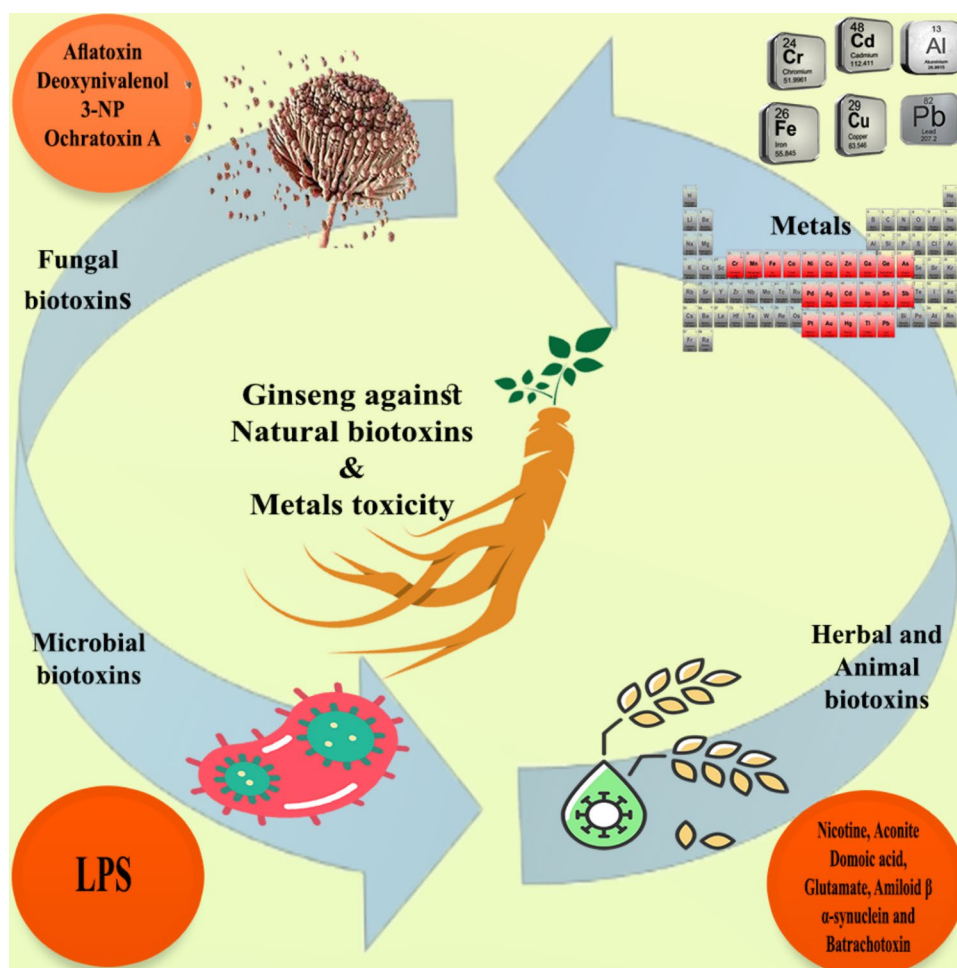
detoxification mechanisms, reduce oxidative stress, and protect against cellular damage caused by toxic substances (El-Demerdash et al. 2021). Moreover, ginseng has been found to modulate the action of various enzymes involved in drug metabolism, potentially influencing the pharmacokinetics and toxicity of certain medications (Ramanathan and Penzak 2017).

We have previously reviewed the potential antidotal effects of various natural substances, including ellagic acid (Naraki et al. 2022), mangosteen (*Garcinia mangostana*) (Mohammadi Zonouz et al. 2023), pomegranate (*Punica granatum*) (Hosseini et al. 2023), glycyrrhiza (Heidari et al. 2021), *Scutellaria baicalensis* (Ahmadi et al. 2022), onion (*Allium cepa*) (Dorrigiv et al. 2021), rosemary (*Rosmarinus officinalis*) (Alavi et al. 2021), garlic (*Allium sativum*) (Dorrigiv et al. 2020), milk thistle (*Silybum marianum*) (Fanoudi et al. 2020), hesperidin (Tabeshpour et al. 2020), lycopene (Hedayati et al. 2019), cinnamon (*Cinnamomum zeylanicum*) (Dorri et al. 2018), turmeric (*Curcuma longa*) (Hosseini and Hosseinzadeh 2018), green tea (*Camellia sinensis*) (Rameshrad et al. 2017), *Berberis vulgaris* (Mohammadzadeh et al. 2017), black Seed (*Nigella*

Sativa) (Tavakkoli et al. 2017), and saffron (*Crocus sativus*) (Razavi and Hosseinzadeh 2015) against natural and chemical toxic agents. In this review, we aim to provide a comprehensive overview of the detoxifying mechanisms of ginseng. Unlike a previous review that focused on the antidotal effects of ginseng monomer saponins in organ toxicity (Wang et al. 2022), we will gather evidence to support the potential preventive and antidote effects of ginseng and its derivatives for various natural toxic agents and metal toxicity and discuss the underlying involved mechanisms (Fig. 3). In a recently published article (Rameshrad et al. 2024), the protective effects of ginseng against chemical toxic agents were discussed. Although the traditional usage of ginseng in toxicities is not clear, its healing effects on the vital organs, in addition to its anti-oxidative, anti-inflammatory, and immune-modulatory properties, make it a promising choice for toxicities.

We will explore the potential therapeutic applications of ginseng as an adjunct to conventional drug therapies, emphasizing the need for further research in this area. This research has the potential to provide valuable insights into the therapeutic applications of ginseng in mitigating the

Fig. 3 *P. ginseng* is a perennial herb with a thick man-root, a 30–60 cm high stem, 3–4 petiolate and palmate-pinnate leaves, and sphere-shaped red berries fruits. This golden herb has protective effects against natural toxic compounds and metal toxicity



harmful effects of toxic substances, contributing to the field of toxicology.

Method

A comprehensive literature search was performed using the keywords ginseng, ginsenoside, gintonin, *Panax ginseng*, protective, antioxidant, natural toxicant, toxic metal, scorpion venom, bee venom, snake venom, botulinum toxin, tetanus toxin, diphtheria toxin, Shiga toxin, enterotoxin, shellfish poisoning, ricin, aflatoxin, deoxynivalenol, three-nitro propionic acid, ochratoxin A, endotoxin, nicotine, aconitine, domoic acid, α -synuclein, amyloid β , glutamate, batrachotoxin, aluminum, cadmium, chrome, copper, iron, and lead in PubMed/Medline, Science Direct, Scopus, and Google Scholar. Additionally, the reference list of the relevant articles was searched, and related in vitro, in vivo, and clinical trial research articles published in English between 2000 and 2024 were included. Studies on the poly-herbal compounds containing *P. ginseng* and studies on the other ginseng species were excluded. The antidote effects of *P. ginseng* and its related compounds were categorized into two main headings: natural toxic agents and metal-induced toxicity. No eligible clinical trials were found.

Results

Natural compounds induced toxicity

P. ginseng and its related compounds decrease the deteriorative effects of several natural toxic agents, including fungal (aflatoxin-B1 (AFB-1), deoxynivalenol, three-nitro propionic acid (3-NP), and ochratoxin A), microbial (lipopolysaccharide (LPS)), herbal and algae produced (nicotine, aconite, and domoic acid), and animal (α -synuclein, amyloid β (A β), glutamate, and batrachotoxin) biotoxins.

Fungal biotoxins

Aflatoxin (AFB-1) AFB-1 was first defined in the early 1960s as a potent hepatotoxic, carcinogenic, mutagenic, and teratogenic toxin in laboratory animals. Its metabolites formed DNA adduct (Rushing and Selim 2019). It has been reported that exposing rats to aflatoxin increases the serum level of ALT, AST, low-density lipoprotein (LDL), and hepatic MDA levels. It induces nodular cirrhosis and fibrosis in animal livers, which are ameliorated by ginseng extract (Abdel-Wahhab et al. 2010). Alcoholic extract of *P. ginseng* putatively has hepatoprotective effects against AFB-1-induced hepatocellular carcinoma. It ameliorates inflammatory cytokines (IL-6 and high sensitive C reactive protein) and modulates

oxidative biochemical markers (MDA, SOD, and nuclear factor (erythroid-derived 2)-like 2 (NFE2L2)). Furthermore, this extract reduces serum alpha-fetoprotein levels, a tumor marker (Aborehab and Waly 2019).

Ginseng extract reduces the negative impacts of AFB-1 on DNA fragmentation, fatty-acid synthesis, and lipid peroxidation in the testis and hepatic cells. This extract enhances the ratio of micronucleated polychromatic erythrocytes to micronucleated normochromic erythrocytes (PCE/NCE) in bone marrow and phospholipid hydroperoxide-glutathione-peroxidase gene expression as well as GSH content in the liver and testis of poisoned animals (Abdel-Aziem et al. 2011). Phospholipid hydroperoxide-glutathione-peroxidase plays a crucial role as a selenium-dependent antioxidant in stabilizing condensed chromatin and safeguarding sperm nuclei against DNA oxidation (Imai and Nakagawa 2003). Yong-Seong et al. showed that Korean red ginseng supplementation in rats prevents AFB-1-induced hepatotoxicity. Korean red ginseng decreased AST, ALT, and MDA levels and enhanced glutathione peroxidase (GPx), superoxide dismutase (SOD), and catalase (CAT) activities (Kim et al. 2011). AFB-1 exposure also induces histopathological alteration, apoptosis, and necrosis in hepatic cells. Also, it leads to inflammatory cell infiltration to the liver, and ginseng ameliorates these changes (Kim et al. 2011). A chronic combination of *P. ginseng* with AFB-1 decreases AFB-1-induced DNA fragmentation and upregulates GPx, SOD1, and CAT gene expression. The lipid peroxidation level is decreased, while glutathione (GSH) content is increased (Hassan et al. 2015). In mice, aflatoxin B increases hepatic expression of matrix metalloproteinase (MMP)-1 and MMP-7 (Al-Mudallal 2023) which are downregulated by ginseng crude extract (Abdulateef et al. 2024).

Deoxynivalenol Deoxynivalenol or vomitoxin is one of the most common mycotoxins produced by *Fusarium* species, which causes harmful effects, such as growth suppression, neuro- and reproductive-toxicities, and impairment of the gastrointestinal tract in both humans and livestock (Yao and Long 2020).

It has been reported that ginsenoside Rb1 improves deoxynivalenol-induced immune damage by improving the histopathological alterations in splenic cells and enhancing its T-lymphocyte subsets (CD4⁺, CD8⁺) content. It improves the levels of IL-2 and -6, interferon (IFN)- γ , TNF- α , and the production of immunoglobulins (Ig) (IgA, IgM, and IgG) reduced by deoxynivalenol (Rajput et al. 2021).

Three-nitropropionic acid (3-NP) 3-NP is a mitochondrial mycotoxin that interferes with ATP synthesis and is commonly used as an agent for experimental models to study Huntington's disease (Ludolph et al. 1991).

It has been shown that -Jun NH2-terminal kinases (JNKs) and p38MAPK induce neurotoxicity in Huntington's disease (Santana-Martínez et al. 2018). NF- κ B is activated in the striatum in Huntington's disease (Santana-Martínez et al. 2018; Haddad 2005). MAPK and NF- κ B are two essential signaling pathways that promote inflammatory signals and induce the expression of TNF- α , IL-1 β , and IL-6 cytokines (Lai et al. 2017; Santana-Martínez et al. 2018). Rg1 induces neuro-protection in 3-NP-induced Huntington's disease by suppressing MAPK and NF- κ B pathways in the striatum (Yang et al. 2021).

Gintonin, another component of ginseng, is a ligand for G protein-coupled lysophosphatidic acid receptor (LPARs) that produces Ca²⁺ second messenger and starts further intracellular communications by activating Ca²⁺-dependent kinases, receptors, and neurotransmitter release (Choi et al. 2015).

Gintonin prevents cell death in the striatum after 3-NPA toxicity probably by the activation of LPARs and nuclear factor erythroid 2-related factor 2 (Nrf2) pathway; further, it may possess inhibitory effects on the NF- κ B signaling pathway (Jang et al. 2019).

The Kelch-like ECH-associated protein 1 (Keap1)-Nrf2 signaling pathway is the most essential cellular antioxidant system that controls the expression of various antioxidant enzymes. Oxidative stress results in the dissociation of this complex and the entrance of Nrf2 into the nucleus, which accelerates the transcription of antioxidant enzymes (Yu and Xiao 2021).

In another study, Rg1 showed protective effects against 3-NP toxicity by inhibiting the NF- κ B and MAPK pathways in the striatum (Yang et al. 2021). Protopanaxatriol is the metabolic product of ginsenoside Rg1 and is considered a neuroprotective ginseng extract. In 3-NPA-induced Huntington's disease, protopanaxatriol activates Nrf2, improves mitochondrial function, enhances mitochondrial succinate dehydrogenase and SOD activity, and increases free radicals scavenging activity (Gao et al. 2015).

In cultured rat striatal neurons, 3-NP elevates intracellular Ca²⁺ content and reduces mitochondrial transmembrane potential; ginseng's saponins restore these toxic effects and prevent striatal neuronal cell deaths (Kim et al. 2005).

It has been shown that 3-NPA-induced cochlear damage is alleviated by Korean Red Ginseng. It probably suppresses the degeneration of fibrocytes in cochlear lateral walls and spiral ganglion cells (Tian et al. 2013).

Ochratoxin A Ochratoxin A is a teratogen mycotoxin that is produced by *Aspergillus ochraceus*. It is a frequent food contaminant and causes well-known toxicities, such as nephrotoxicity and hepatotoxicity (Khoi et al. 2021).

Ochratoxin A results in tubular and glomerular degeneration and induces interstitial and intraglomerular hemorrhage. It decreases DNA, protein, and glycogen content and

decreases alkaline phosphatase activity in the kidney cells. Ginseng ameliorates histopathological, histochemical, and morphometric kidney injuries of ochratoxin A (Morsy et al. 2012).

Microbial biotoxins

Lipopolysaccharide (LPS) LPS or endotoxin, a constituent of the outer membrane of gram-negative bacteria, is responsible for the inflammatory response syndrome (De Punder and Pruimboom 2015). Previous investigation indicated that LPS upregulates the expression of pro-inflammatory agents, including IL-6, IL-1 β , and IL-8, which are remarkably downregulated by Rb3 treatment. Rb3 significantly suppresses the phosphorylation of p38 MAPK and p65 NF- κ B and diminishes the expression of total AKT as a critical inflammatory response pathway of periodontal tissues to *Porphyromonas gingivalis* LPS. It downregulates the expression of TLR-2 and ameliorates alveolar bone resorption in experimental periodontitis rats induced with *P. gingivalis* LPS (Sun et al. 2020). Further, ginsenoside Rg6 relieves *P. gingivalis* LPS-induced inflammation and periodontitis (Lee et al. 2023). Ginsenoside Rg1 inhibits the reactive oxygen species (ROS)/nucleotide oligomerization domain-like receptor protein (NLRP)3 inflammasome signaling pathway and ameliorates LPS-induced endometritis (Song et al. 2023). Additionally, by activating Nrf2 signaling and inhibiting inflammasomes, ginsenosides Rg1 alleviates the deteriorative effects of LPS in the liver (Zhou et al. 2024), kidney (Ji et al. 2024), and CNS (Kong et al. 2024). In LPS-induced acute kidney injury in mice, ginsenoside Rg1 possessed nephroprotective effects by upregulating SIRT1 expression and inactivating NF- κ B signaling (Hu et al. 2024).

In mice, LPS-induced lung inflammation was attenuated by inhaled panaxadiol. In vitro studies showed panaxadiol reduced epithelial apoptosis by inhibiting TNF- α /TNF α R and IL-7/IL-7R (Wang et al. 2024). LPS-induced inflammation in human dental pulp cells was attenuated by ginsenoside Rb1. It reduced inflammatory cytokines and suppressed the expression of ICAM-1 and VCAM, inhibited PI3K/Akt, and decreased phosphorylated I κ Ba, p65, ERK, and JNK (Nam et al. 2024). However, the role of the PI3K/Akt pathway in NF- κ B regulation and inflammation is controversial (He et al. 2022).

LPS-induced platelet activation and platelet-leukocyte and platelet-neutrophil aggregation are inhibited by red ginseng extract. It also reduced the plasma level of tissue factor and platelet factor 4 in septic mice (Lee et al. 2024).

Herbal and algae-produced biotoxins

Nicotine Nicotine, a highly toxic alkaloid found in tobacco products, is a potentially addictive substance (Siqueira et al. 2017). Nicotine administration in rats represses the secretion of free testosterone, luteinizing- and follicle-stimulating hormones (LH and FSH), and the levels of expression of the genes encoding these hormones. Nicotine consumption increases DNA damage in the testicular tissue and causes sperm abnormalities. However, treatment of male rats with *P. ginseng* or *P. ginseng* nanoparticles prevents the reproductive toxicity induced by nicotine (Linjawi 2015).

Aconite Aconitine, the main active component of Aconitum plants, has therapeutic benefits but also poses risks, particularly cardiotoxicity and neurotoxicity, which limits its clinical use (Li et al. 2022). The protective effects of ginsenoside Rb1 were explored against aconitine's cardiotoxicity. Rb1 demonstrated cardiovascular protection by inhibiting pathological damage, reducing myocardial enzyme levels, and balancing apoptotic protein expression. Additionally, Rb1 mitigated the arrhythmic effects of aconitine in rats by normalizing action potential conduction and calcium signaling, thus offering a potential method to reduce the cardiotoxicity associated with aconitine-containing medicines (Zhang et al. 2024). One study evaluated the rationality and compatibility of Shenfu Formula (SFF), a typical Chinese medicine composed of *P. ginseng* and *Aconitum carmichaelii*. Using Caco-2 cells, the study investigated the permeability of *A. carmichaelii* marker compounds when combined with *P. ginseng*. Notably, *P. ginseng* inhibited the intestinal absorption of highly toxic compounds including aconitine, mesaconitine, and hyaconitine from *A. carmichaelii*. P-glycoprotein (P-gp) and breast cancer resistance protein (BCRP) were involved in this process. Additionally, *P. ginseng* induced P-gp activity, facilitating the efflux of toxic *A. carmichaelii* alkaloids while preserving other active components. Future studies will explore the mechanism behind the reduced toxicity when combining *A. carmichaelii* with *P. ginseng*, guiding formula optimization (You et al. 2015). Another study focused on ginsenoside Rg1 and three representative alkaloids from aconite: aconitine (AC), benzoyl-aconine (BAC), and aconine (ACN). The aim was to investigate how ginsenoside Rg1 affects the pharmacokinetics of these alkaloids and provide evidence for their compatibility mechanism. In rat experiments, ginsenoside Rg1 altered the pharmacokinetic parameters of AC and BAC: peak concentration and area under the concentration–time curve were affected. Further in vitro investigations using Caco-2 cell monolayers and rat liver microsomes revealed that ginsenoside Rg1 promoted AC and BAC absorption while accelerating AC metabolism. These findings suggest that ginsenoside Rg1 may reduce aconite toxicity and enhance its efficacy by

influencing alkaloid absorption and metabolism (Xu et al. 2020).

Domoic acid Domoic acid is a naturally occurring biotoxin usually produced by the microscopic algae *Pseudo-nitzschia* and is the cause of amnesic shellfish poisoning (Jeffery et al. 2004). A study investigated the potential effects of ginsenosides from *P. ginseng* in countering amnesic shellfish poisoning, specifically domoic acid–induced excitotoxicity. Mice were orally administered *P. ginseng* at two different dosages (50 and 100 mg/kg) over 28 days. Domoic acid was used to induce excitotoxicity, with piracetam-injected animals serving as positive controls. *P. ginseng* supplementation successfully reversed domoic acid–induced cognitive impairment. Additionally, *P. ginseng* restored GABA levels and antioxidant enzymes and changes in BDNF, NF- κ B, and apoptotic markers (Bcl₂, Bax, and caspase-3). The study sheds light on oxido-inflammatory responses and protecting against domoic acid–induced neuronal impairments. These findings have practical implications for pharmacological applications of *P. ginseng* in managing domoic acid–induced excitotoxicity (Ramya et al. 2022).

Animal biotoxins

α -Synuclein Research suggests that α -synuclein aggregation is key in Parkinson's disease (PD) pathogenesis and its toxicity might directly disrupt mitochondrial homeostasis (Stefanis 2012). A study focused on ginsenosides' potential showed that Rb1 was an effective inhibitor of α -synuclein fibrillation and toxicity, capable of disaggregating preformed fibrils and preventing further polymerization. Rb1 also stabilized non-toxic, soluble oligomers without β -sheet content, suggesting its mechanism of action (Ardah et al. 2015).

Amyloid β (A β) A β aggregation in the brain tissue is reflected as an aspect of the potential pathogenesis of Alzheimer's disease (Khandelwal et al. 2011).

A previous study suggested that ginsenoside Rg1 diminishes brain A β content in an Alzheimer's disease model by downregulating cyclin-dependent kinase 5 (CDK5) expression. This event inhibits peroxisome proliferator–activated receptor γ (PPAR γ) phosphorylation at Ser 273, modulates PPAR γ targets genes, increases the expression of insulin-degrading enzyme, and decreases the β -amyloid cleavage enzyme 1 (BACE1) expression level. Ginsenoside Rg1 decreases neuronal apoptosis and downregulates BACE1 and amyloid precursor protein expression (Quan et al. 2020).

It has been reported that ginsenoside Rg2 can relieve the impaired memory induced by A β_{25-35} in rats. It enhances the B-cell lymphoma 2/Bcl-2-associated X protein (Bcl-2/Bax) ratio, attenuates the cleavage of caspase-3, and ameliorates hippocampal neuron impairment via elevating the

phosphatidylinositol 3-kinase/protein kinase B (PI3K/Akt) signaling pathway (Cui et al. 2021).

Choi et al. investigation showed that premedication of human neuronal cells exposed to A β with Korean red ginseng increases cell viability and decreases apoptotic factors (caspase-3 activity and Bax/Bcl-2 ratio). Korean red ginseng suppresses A β -induced mitochondrial dysfunction. It inhibits NF- κ B activation and proapoptotic gene Nucling expression in A β -treated cells (Choi et al. 2020). Furthermore, co-treatment of PC12 cells with red ginseng oil compositions (linoleic acid, β -sitosterol, and stigmasterol) reduces A β induced oxidative stress and apoptotic responses by modulating the mitochondrial membrane potential, intracellular Ca²⁺ content, Bax/Bcl-2 ratio, and the level of caspase-9, caspase-3, and caspase-8. This oil reduces pro-inflammatory mediators inducible NOS (iNOS), interleukin (IL)-6, tumor necrosis factor- α (TNF- α), and COX-2 through downregulation of NF- κ B and MAPK pathways (Lee et al. 2018). MAPKs are essential in apoptosis (Cagnol and Chambard 2010).

Glutamate Glutamate, an excitatory neurotransmitter in the CNS, is commonly used as a neurotoxic agent in inducing experimental chronic neurodegenerative ailments such as Alzheimer's disease (Hynd et al. 2004).

A previous study reported that premedication of neurons with Rg1 prevents glutamate-induced endoplasmic reticulum stress and neurotoxicity by blocking the PKR-like endoplasmic reticulum kinase (PERK)-eukaryotic translation initiation factor 2 (eIF2 α)-activating transcription factor 4 (ATF4) signaling pathway. It increases cell viability and decreases apoptosis (Gu et al. 2020). PERK-eIF2- α -ATF4 pathway is one of several apoptosis mediators in cell death related to endoplasmic reticulum stress in brain ischemia/reperfusion (I/R) injury (Xin et al. 2014).

20(S)-Protopanaxadiol pretreatment protects against glutamate-induced excitotoxicity in PC12 cells. This premedication increases cell viability and decreases apoptosis. It decreases cytosolic and mitochondrial reactive oxygen species (ROS) production and increases the number of regular and functional mitochondria (Bak et al. 2016). The activation of the Nrf2/ heme oxygenase-1 (HO-1) pathway might be related to the anti-oxidative activities of ginseng oligosaccharides against glutamate-induced neuron injury (Yan et al. 2022).

Batrachotoxin Batrachotoxin is one of the most potent neuro- and cardiotoxic agents found in some species of birds and frogs (Cataldi 2010). Structurally, it is a steroidal alkaloid and persistently activates Na⁺ channels (Catterall et al. 2007). In the mouse brain, Duan et al. showed 20(S)protopanaxadiol, 20(S)protopanaxatriol, and Rh 2 could block the binding of [3H]batrachotoxinin A 20- α -benzoate to

Na⁺ channels through batrachotoxin binding site (Duan et al. 2006). Using site-directed mutagenesis, Lee et al. (Lee et al. 2008) proved the findings of the Duan et al. study (Duan et al. 2006). They showed the fundamental role of batrachotoxin binding sites in N418 and L421 residues of the NaV1.2 and the L437 residue of NaV1.4 channels in ginsenoside Rg-3-mediated Na⁺ channel inhibition (Lee et al. 2008).

Based on the gathered data, we showed the antidote/protective properties of ginseng and its main components against a wide range of natural toxic agents (Table 1). The activation of PI3K/Akt, LPARs, and Nrf2 pathways, the deactivation of MAPKs and NF- κ B signaling pathways, and the downregulation of TLR-2 expression are important in their anti-oxidative and anti-inflammatory properties. Some of these herbal materials block both intrinsic and extrinsic apoptotic pathways. They inhibit endoplasmic reticulum stress by blocking the PERK-eIF2- α -ATF4 pathway. Some decline oxidative biomarkers, suppress tumor markers, stabilize condensed chromatin, and protect against DNA oxidation. Further, some may influence the biotoxin pharmacokinetic, absorption, and metabolism. We may ask whether ginseng and its main component could be evaluated in batrachotoxin toxicity (Fig. 4).

Metals

P. ginseng and its related compounds decrease the deteriorative effects of metal toxicity, including aluminum (Al), cadmium (Cd), chrome (Cr), copper (Cu), iron, and lead (Pb).

Aluminum (Al)

Al can easily enter the human body and has undeniable human health. In cases of acute toxicity, it can lead to neurotoxicity, which can be treated with deferoxamine and/or hemodialysis (Hoffman et al. 2023a). Chronic toxicity can cause damage to multiple organs including the respiratory, nervous, reproductive, cardiovascular, endocrine, hepatobiliary, kidney, hematopoietic, musculoskeletal, and immune systems. The toxic effects of Al may be associated with oxidative stress, inflammation, immunosuppression, metabolic impairment, inhibition of bone formation and mineralization, genotoxicity, and amyloidogenic and anti-amyloidolytic activity leading to cell degeneration, apoptosis, necrosis, atrophy, and dysplasia. Laboratory tests to evaluate Al toxicity include measuring levels of Al in blood, bone, urine, and feces. It is important to reduce Al intake and its absorption in the intestines and skin to prevent and treat aluminum toxicity. Deferoxamine is the main chelator used in Al toxicity treatment but may cause anaphylaxis. Malic, malonic, citric, oxalic, and succinic acids may also increase the excretion of Al through urine. Antioxidants such

Table 1 Summary of the protective effects of *Panax ginseng* and its main components against natural toxic agents

Toxic agent	Dose, duration, and route of exposure to a toxin	Ginseng derivatives, dose, duration, route of exposure	Experimental model	Results	References
AFB	17 µg/kg/day for first 2 weeks, p.o. and fed fumonisin-contaminated diet (250 mg/kg diet) during the 6th to 8th weeks	Ginseng extract (250 mg/kg), p.o. - During the mycotoxin protocol until week 10 - For 2 weeks before AFB-1 administration until the end of fumonisin treatment -For 4 weeks after the toxin protocol stopped	AFB-1 and fumonisin-induced hepatic pre-cancerous lesions in female Sprague–Dawley rats	During, before, or after the treatment: ↓ ALT, AST, MDA, LDL Normalized HDL and cholesterol and TG ↓ Fibrous and damaged tissues in hepatocytes	(Abdel-Wahhab et al. 2010)
		150 µg/kg/day for 3 days, i.p	AFB-1-induced hepatocellular carcinoma in male albino rats	↓ Serum Alpha-fetoprotein level ↓ Serum IL6 ↓ High sensitive C reactive protein ↓ Hepatic MDA ↑ Hepatic SOD ↑ Tissue rat nuclear factor erythroid-derived 2 like 2 protein	
	2.5 mg/kg in diet for 30 days, p.o	KRG (20 mg/kg) for 30 days, p.o	Aflatoxin-induced liver and testis damage in male Sprague–Dawley rats	↓ Micronucleated normochromatic erythrocytes, DNA fragmentation, fatty-acid synthesis mRNA expression ↑ GSH and ↓ LPO ↑ Expression of phospholipid hydroperoxide-glutathione peroxidase mRNA	(Abdel-Aziem et al. 2011)
	150 µg/kg for the last 3 days of KRG treatment, i.p	KRG (250 mg/kg) for 4 weeks, p.o	Aflatoxin B ₁ -induced hepatotoxicity in male Sprague–Dawley rats	↓ AST, ALT and MDA levels ↑ SOD, CAT, and GPx activities ↓ Necrosis and apoptosis in hepatocyte	(Kim et al. 2011)
Deoxynivalenol	80 µg/kg/day, for 12 weeks, p.o	<i>P. ginseng</i> extract (20 mg/kg for 12 weeks, p.o.)	Sprague–Dawley female rats	↓ Liver and kidney MDA ↑ Liver GSH ↑ Liver gene expression of GPx, SOD1 and CAT ↓ Liver DNA fragmentation	(Hassan et al. 2015)
	6 mg/kg (60% of pure aflatoxin B ₁), twice a week, for 2 weeks, p.o	<i>P. ginseng</i> aqueous extract (50, 100, and 150 mg/kg/day for 3 weeks, p.o.)	Aflatoxin-induced spleen, liver and kidney damage in male Swiss albino mice	↓ MMP 1 and 7 expression in the spleen, liver, and kidney	(Abdulateef et al. 2024)
	3 mg/kg for 15 day, p.o	Rb1 (50 mg/kg) for 15 days, p.o	Deoxynivalenol-induced immune injury in C57BL/6 mice	↑ Average daily gain and average daily feed intake ↓ IL-2, IL-6, IFN-γ, and TNF-α, Bax, caspase-3, caspase-9 ↑ SOD, CAT, and GPx Bcl-2	(Rajput et al. 2021)

Table 1 (continued)

Toxic agent	Dose, duration, and route of exposure to a toxin	Ginseng derivatives, dose, duration, route of exposure	Experimental model	Results	References
3-nitropropionic acid	6 mg/kg BID on the first day and 80 mg/kg BID from 2nd to the 4th day, i.p	Rg1 (10, 20, 40 mg/kg), 30 min before 3-NP administration, p.o	3-Nitropropionic acid-induced Huntington's disease in male C57BL/6 mice	↓ Body weight loss ↓ Motor behavior defects ↓ Striatal neuronal loss ↓ Production of pro-inflammatory cytokines in the striatal ↓ Activation of the MAPK and NF-κB pathways in the striatal	(Yang et al. 2021)
	Twice daily for 2 days at 12-h intervals, i.p 60 mg/kg on the first day and 80 mg/kg on the second day	Gintonin (GT) 25, 50 and 100 mg/kg/day, p.o, from 5 days and 1 h before and after the first NPA administration	3-Nitropropionic acid-induced striatal toxicity in male C57BL/6NTac mice	- Improved neurological behavior ↓ Striatal cell death ↓ IL-1β, IL-6, TNF-α, COX-2, and iNOS - Regulated Keap1-Nrf2 ↑ NQO-1, γ-GCLs, HO-1	(Jang et al. 2019)
	60 mg/kg [the 1st day] and 80 mg/kg 3-NP [the 2nd–4nd day], twice daily, i.p	Ginsenoside Rg1 (40 mg/kg for 5 days, p.o.)	3-Nitropropionic acid-induced Huntington's disease in male C57BL/6 mice	↓ MAPKs, NF-κB, and caspase 3 - Alleviated body weight loss, neuronal loss and behavioral defects	(Yang et al. 2021)
	20 mg/kg 3-NP (ip), on d 1, and then 15 mg/kg/d for the next 4 days	Protopanaxatriol (20 mg/kg) p.o., for 5 day	3-Nitropropionic acid-induced oxidative stress in Huntington's disease in male Sprague–Dawley rats	- Alleviate body weight loss ↓ Behavior disorder ↓ Overproduction of ROS in neurons ↓ Hsp70 (has cytoprotective function and induced in neurodegenerative disorders) expression in the striatum ↑ Mitochondrial succinate dehydrogenase activity - Scavenge the hydroxyl radical - scavenge the superoxide anion radicals ↑ Nrf2 entry to the nucleus in the striata ↑ HO-1 and NQO1 protein levels in the striata	(Gao et al. 2015)
	500 mM, intratympanic	KRG, 300 mg/kg/day, oral, 7 days before 3-NPA administration and continued for 7 days after that	3-Nitropropionic acid-induced cochlear damage in male BALB/c mice	↓ Auditory thresholds at both 16 and 32 kHz frequencies post 1 week and post 1 month ↑ Type II fibrocyte count in the cochlear lateral walls ↑ Mean cell counts in spiral ganglion	(Tian et al. 2013)

Table 1 (continued)

Toxic agent	Dose, duration, and route of exposure to a toxin	Ginseng derivatives, dose, duration, route of exposure	Experimental model	Results	References
Ochratoxin A	Ochratoxin A (0.85 mg/kg), i.p., for 2, 4 and 8 weeks	Ginseng (20 mg/kg), i.p., 10 min before OchrA	Ochratoxin-induced toxicity in the kidney in male albino rats	↓ Tubular and glomerular degeneration ↓ Interstitial and intraglomerular hemorrhage ↑ Glomerular area and damage area in kidney cells ↓ Hyaline casts and vacuolar degeneration ↑ In DNA, protein and glycogen content, and alkaline phosphatase activity in the kidney cells	(Morsy et al. 2012)
LPS	1 µg/mL; 24 h	(0, 50, 100, 200, and 400 µM) with or without LPS	LPS-stimulated human periodontal ligament cells (HPLCs)	↓ IL-1β, IL-6, and IL-8 ↓ Phosphorylation of p38 MAPK ↓ p65 NF-κB ↓ Expression of total AKT ↓ In alveolar bone resorption and osteoclast generation	(Sun et al. 2020)
	1 mg/mL injected around the neck on odd days for 20 days	G-Rb3 100 µM, i.p for 20 days	Experimental periodontitis in male Sprague–Dawley rats	Improved kidney pathological findings	(Ji et al. 2024)
	200 µg/kg, i.p, SD	Rg1 (5, 10, and 20 mg/kg/day), gavage, for 21 days	LPS-induced chronic kidney renal fibrosis in male C57BL/6 J mice	↓ Renal fibrosis ↓ Serum creatinine ↓ Blood urea nitrogen ↓ Urinary albumin ↓ Bax ↓ Cleaved caspase-3 ↑ Bcl-2 ↓ β-Gal ↓ ROS ↓ TGF-1β ↓ p-Smad2/3 /Smad2/3 ↑ pNrf2/Nrf2 ↑ HO-1 ↓ AIM2 ↓ caspase-1 ↓ IL-6 ↓ IL-1β	

Table 1 (continued)

Toxic agent	Dose, duration, and route of exposure to a toxin	Ginseng derivatives, dose, duration, route of exposure	Experimental model	Results	References
	10 µg/ml for 24 h	Rg1 (10 µM) for 24 h with/without ML385 (a Nrf2 inhibitor)	Human mesangial cells (specialized cells in the kidney)	↑ HO-1 ↓ AIM2 ↓ caspase-1 ↓ ASC ↓ IL-6 ↓ IL-1β ↓ TGF-1β ↓ p-Smad2/3 /Smad2/3 ML385 reversed all of these aforementioned results	
	200 µg/kg for 21 days, i.p	Rg1 (5, 10, and 20 mg/kg/day, for 21 days, gavage	LPS-induced chronic liver damage in C57BL/6 J male mice	Improved hepatic pathological findings ↓ Hepatic glycogen deposition ↓ Hepatic fibrosis ↓ Serum ALT ↓ Serum AST ↓ Serum IL-6 ↓ Serum IL-1β ↓ Serum TNF-α ↑ pNrf2 ↓ NLRP3 ↓ NLRP1 ↓ AIM2 ↓ Caspase-1 ↓ Bax ↑ Bcl-2	(Zhou et al. 2024)
	10 µg/ml for 24 h	Rg1 10 µM, 24 h, with/without ML385 (a Nrf2 inhibitor)	HepG2	↓ ROS ↑ Nuclear Nrf2 ↑ HO-1 ↑ NQO1 ML385 reversed all of these aforementioned results	
	200 µg/kg for 21 days, i.p	Rg1 (5, 10, and 20 mg/kg/day, for 21 days, gavage	LPS-induced neuroinflammation and cognitive deficits in C57BL/6 J male mice	Improved behavioral dysfunction ↓ Neuronal damage ↓ ROS ↓ AIM2 inflammasome ↓ IL-6 ↓ IL-1β ↓ Caspase-1 ↓ Ferroptosis ↑ Nrf2 ↑ HO-1 ↑ NQO1	(Kong et al. 2024)

Table 1 (continued)

Toxic agent	Dose, duration, and route of exposure to a toxin	Ginseng derivatives, dose, duration, route of exposure	Experimental model	Results	References
	10 µg/ml for 24 h	Rg1 10 µM, 24 h, with/without ML385 (a Nrf2 inhibitor)	HT22	↓ ROS ↓ IL-6 ↓ IL-1β ↓ Caspase-1 ↑ Nrf2 ↑ HO-1 ↑ NQO1 ML385 reversed all of these aforementioned results, reduced anti-ferroptosis effect of Rg1	(Nam et al. 2024)
	1 µg/ml for 24 h	Rb1 (1, 5, and 10 µM) for 24 h	LPS-induced human dental pulp cells inflammation	↓ TNF-α ↓ IL-6, IL-8, and IL-1β ↓ VCAM ↓ ICAM ↓ PI3K/Akt ↓ p-IkB ↓ p-p65 ↑ Cytosolic p-p65 ↓ Nucleus p65 ↓ p-erk ↓ p-p38 ↓ p-INK	
	15 mg/kg, i.p, SD	Red ginseng extract (100, 200, and 400 mg/kg), gavage, for 7 days	Sepsis-induced platelet-leukocyte aggregation in ICR male mice	↓ Platelet-leukocyte aggregation ↓ Platelet-neutrophil aggregation ↓ Platelet activation ↓ Fibrin formation ↓ Tissue factor ↓ Platelet factor 4	(Lee et al. 2024)
Nicotine	1.5 mg/kg for 4 weeks, p.o	<i>P. ginseng</i> (20 mg/kg) for 4 weeks, p.o	Nicotine-induced reproductive disorders in male albino rats	Regulated testosterone, LH, and FSH levels and improved DNA damage in testicular tissue	(Linjawi 2015)

Table 1 (continued)

Toxic agent	Dose, duration, and route of exposure to a toxin	Ginseng derivatives, dose, duration, route of exposure	Experimental model	Results	References
Domoic acid	3 mg/kg, i.p., SD	Ginsenosides (50 and 100 mg/kg) for 28 days, p.o	Domoic acid-induced excitotoxicity in balb/c mice	↓ Cognitive impairment ↑ SOD ↑ CAT ↑ GPx ↑ Glutathione reductase ↓ MDA ↓ Protein oxidation ↑ GABA ↓ MAPK8 ↑ CREB ↑ BDNF ↓ Bax ↓ Caspase-3 ↑ Cleaved caspase-3 ↓ NF-κB	(Ranya et al. 2022)
α-Synuclein	Aged α-synuclein (0.5, 1, and 5 μM), for 48 h	Rb1, Rg1, or Rg3 (0.5, 1, and 5 μM)	α-Synuclein-induced cytotoxicity in BE(2)-M17 human neuroblastoma cells	Rb1: ↓ α-Synuclein fibrillation and toxicity ↓ Seeded polymerization of α-synuclein disaggregate preformed fibrils	(Ardah et al. 2015)
Amyloid beta	8 μM for 24 h	60 μM ginsenoside Rg1 1 h before exposure	A neuronal model of Alzheimer's disease in hippocampal neurons of 2-day-old Sprague–Dawley rats	↓ β-amyloid cleavage enzyme 1 and amyloid precursor protein gene expression and apoptosis ↓ cdk5 protein and mRNA expression ↓ PPARγ phosphorylation	(Quan et al. 2020)
	5 μg injected bilaterally into the hippocampal region of the rat on the 15th day after Rg2 beginning	Rg2 (25–100 mg/kg) for 6 weeks, p.o	Aβ-induced neurotoxicity in male Sprague–Dawley rats	↑ Bcl-2/Bax ratio PI3K/Akt signaling pathway ↓ caspase-3 Improved memory impairment and histopathological injury in hippocampus	(Cui et al. 2021)
	5 μmol/L treated for 24 h	20 μg/mL, 1 h prior to exposure	Aβ-induced apoptosis in human neuronal cells	↑ Bcl-2/Bax ratio ↓ ROS and NF-κB	(Choi et al. 2020)
	50 μM for 24 h	Red ginseng oil extract 20 μM	Aβ _{25–35} -stimulated PC12 cells	↓ iNOS, TNF-α, IL-6 and COX-2 ↓ Ca ²⁺ , Bax/Bcl-2 ratio, and caspase-9, caspase-3, and caspase-8	(Lee et al. 2018)
Glutamate	5 μM GLU for 30 min	Pretreatment with 20 μM Rg1 for 24 h	Neuronal culture from the cortices of 17-day-old Wistar rat fetuses	Reduced apoptosis by inhibiting the PERK-ATF4-CHOP pathway	(Gu et al. 2020)

Table 1 (continued)

Toxic agent	Dose, duration, and route of exposure to a toxin	Ginseng derivatives, dose, duration, route of exposure	Experimental model	Results	References
	5 mM glutamate for 24 h	Pre-treated with 5, 10, or 20 µM PPD	Glutamate-induced mitochondrial dysfunction in PC12 cell line	Inhibiting the nuclear condensation and suppressing the number of Annexin V-positive cells ↑ cell viability ↓ Apoptosis ↓ Bax and caspase-3 ↑ Bcl-2 expression ↓ ROS production Improving mitochondrial shape and function ↓ Histological damages	(Bak et al. 2016)

Aβ amyloid β, *AFB-1* aflatoxin-B1, *AIM2 inflammasome*, absent in melanoma 2 inflammasome, *ALT* alanine aminotransferase, *AST* aspartate aminotransferase, *ATF4* activating transcription factor 4, *Bax* Bcl-2-associated X protein, *Bcl-2* B-cell lymphoma 2, *BDNF* brain-derived neurotrophic factor, *BID* twice a day, *CAT* catalase, *CHOP* CCAAT/enhancer-binding protein homologous protein, *COX-2* cyclooxygenase-2, *CREB* cAMP response element-binding protein, *FSH* follicle-stimulating hormone, *GABA* gamma-aminobutyric acid, *γ-GCLs* γ-glutamyl cysteine ligase regulatory subunit, *GPx* glutathione peroxidase, *GSH* glutathione, *HDL* high-density lipoproteins, *HO-1* hemeoxygenase-1, *Hsp 70* heat shock protein70, *IFN-γ* interferon gamma, *IL-1* interleukin-1, *iNOS* inducible nitric oxide synthase, *i.p.* intraperitoneal, *KRG* Korean red Ginseng, *LDL* low density lipoproteins, *LH* luteinizing hormone, *LPO* lipid peroxidation, *MAPK* mitogen-activated protein kinases, *MDA* malondialdehyde, *MMP* matrix metalloproteinase, *NF-κB* nuclear factor kappa-light-chain enhancer of activated B cells, *NLRP* nucleotide oligomerization domain-like receptor protein, *NPA* 3-nitropropionic acid, *Nrf2* nuclear erythroid2-related factor2, *NQO1* nicotinamide-adenine dinucleotide phosphate:quinone oxidoreductase-1, *PERK* PKR-like endoplasmic reticulum kinase, *p.o.* per oral, *PI3K/Akt* phosphatidylinositol 3-kinase/protein kinase B, *PPARα/γ* peroxisome proliferator-activated receptor alpha/gamma, *ROS* reactive oxygen species, *Sirt1* sirtuin-1, *SOD* superoxide, *TG* triglycerides, *TNF-α* tumor necrosis factor-α

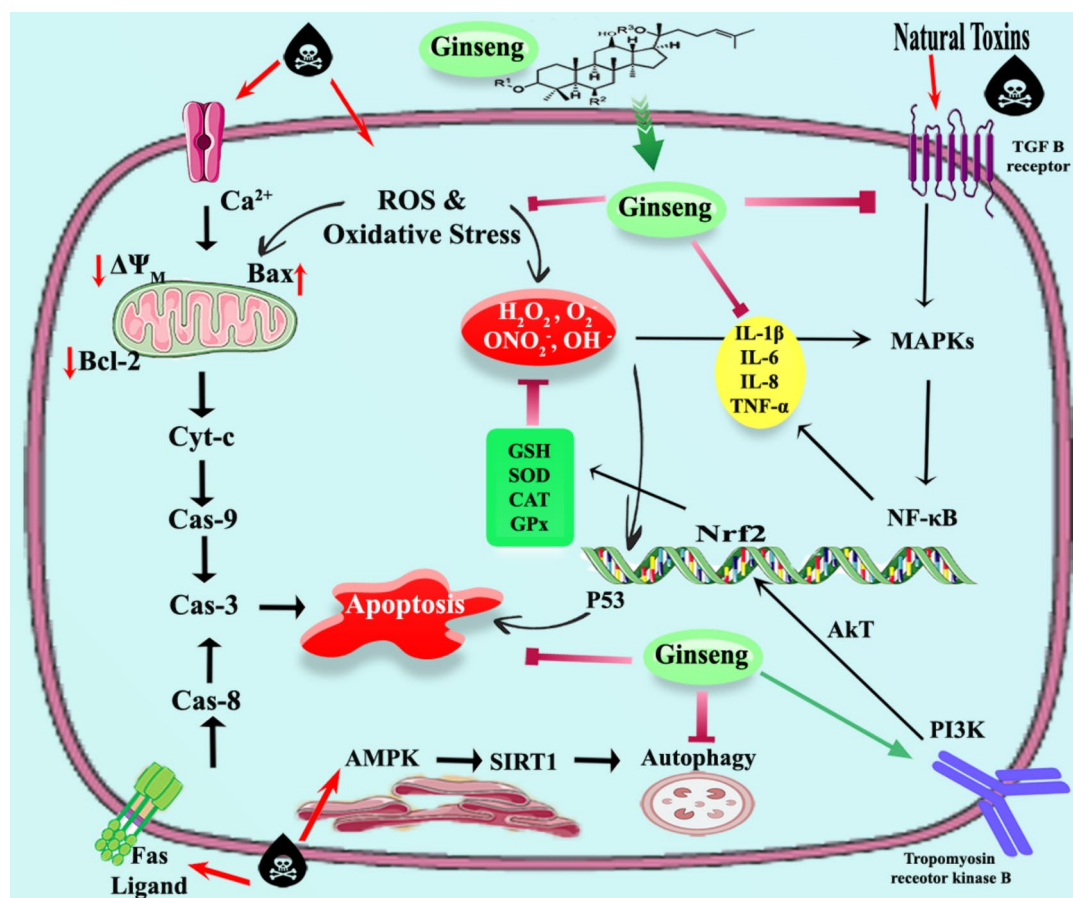


Fig. 4 The molecular and cellular protective mechanisms of ginseng and its main constituents against natural toxins. Akt, protein kinase B; AMPK, AMP-activated protein kinase; Bax, Bcl-2-associated X protein; Bcl-2, B-cell lymphoma 2; cas-, caspase; CAT, catalase; cyt-c, cytochrome-c; GPx, glutathione peroxidase; GSH, glutathione; HO-1, heme oxygenase; IL-, interleukin; MAPK, mitogen-activated

protein kinase; NF-κB, nuclear factor κ-light-chain enhancer of activated B cells; Nrf2, nuclear factor erythroid 2-related factor 2; PI3K, phosphatidylinositol 3-kinase; ROS, reactive oxygen species; Sirt1, sirtuin 1; SOD, superoxide dismutase; TGF-β1, transforming growth factor-β1; TNF-α, tumor necrosis factor-α

as selenium, melatonin, boric acid, vitamin C, and certain biologically active compounds are suggested to cope with the free radicals and related deleterious effects induced by aluminum toxicity (Igbokwe et al. 2019).

Brain-derived neurotrophic factor (BDNF) and fibroblast growth factor (FGF) connecting the PI3K/Akt signaling pathway enhance Bcl-2 transcription and inhibit cell-related apoptosis pathways (Zhong et al. 2019). In the D-galactose and AlCl_3 -induced aging model, Rg1 administration ameliorates cognitive deficits. This ginsenoside restores the FGF-Akt, BDNF-tropomyosin receptor kinase B, and Bcl-2 signaling pathways in the hippocampus and prefrontal cortex. Additionally, it inhibits caspase-3 activation and neuronal apoptosis (Zhong et al. 2020). Also, the neuroprotective effects of ginseng protein in this model have been proven with the activation of cyclic adenosine monophosphate/protein kinase A/cAMP response element-binding protein

(cAMP/PKA/CREB) (Li et al. 2017) and PI3K/Akt pathways (Li et al. 2016).

Ginseng ethanol extract provides neuroprotective effects against AlCl_3 -induced Alzheimer's disease (Abdelghany et al. 2019). Restoring the dysfunction of various neurotransmitters could be a possible mechanism for the neuroprotective effects of ginsenosides in this model (Zhang et al. 2016).

Rg3 is effective against AlCl_3 -induced osteoporosis. This effect is achieved by decreasing Al accumulation and attenuating oxidative stress. It facilitates bone formation, inhibits bone resorption, and promotes the mRNA expression of growth regulation factors (Song et al. 2020).

Ginsenoside Rb1 increases the expression of osteoblast growth regulation factors, including transforming growth factor-beta1 (TGF-β1), bone morphogenetic protein-2, insulin-like growth factor I, and core-binding factor α1. This event assists osteogenesis and protects against

AlCl₃-induced osteoblast dysfunction. Further, it attenuates AlCl₃-induced lesions in osteoblast histology by suppressing oxidative stress (Zhu et al. 2016).

Post therapy with ginsenoside Rb1 diminishes tau phosphorylation by decreasing phosphorylated glycogen synthase kinase-3 (p-GSK3) and increasing protein phosphatases 2A (PP2A) level in the brain; it improves learning and memory dysfunction induced by AlCl₃ (Zhao et al. 2013). GSK-3 phosphorylates tau (Hanger et al. 1992) and GSK-3 phosphorylation maintains its activity in neurons (Forde and Dale 2007). PP2A is a major tau phosphatase (Liu et al. 2005).

Cadmium (Cd)

Cd is a commonly used metal in industrial fields such as electric batteries and electronic components and has been listed as a human carcinogen (Bertin and Averbeck 2006). Urinary cadmium concentration and ancillary tests, including pulse oximetry and chest radiography, are promising in Cd toxicity diagnosis. However, in acute toxicities, history, physical examination, and symptoms are more useful than cadmium concentration. Cd toxicities cause multiorgan damage, such as GI problems and pneumonitis in acute exposure, and proteinuria, nephrolithiasis, osteomalacia, and cancer in chronic exposure (Hoffman et al. 2023b). Its toxic effects are related to oxidative stress, generation of reactive oxygen species, and damage to proteins, lipids, and DNA. Cadmium toxicity decreases cellular antioxidant defenses. In mitochondria, the main target of Cd toxicity, it induces mtDNA mutation, alters gene expression, inhibits respiratory chain complexes II and III, reduces ATP synthesis, alters the inner mitochondrial permeability, activates caspase-8 and caspase-9, and leads to apoptosis (Genchi et al. 2020). In acute pulmonary exposure, corticosteroids are recommended (Hoffman et al. 2023b). In acute oral exposure, emesis or gastric lavage reduces Cd intake (Genchi et al. 2020; Hoffman et al. 2023b). Chelating therapy using chelating agents such as ethylenediaminetetraacetic acid (EDTA) (Genchi et al. 2020), British anti-Lewisite (Genchi et al. 2020), dimercaprol (BAL) (Genchi et al. 2020), 2,3-dimercaptosuccinic acid (DMSA or succimer) (Genchi et al. 2020; Hoffman et al. 2023b), 2,3-dimercapto-1-propane sulfonic acid (DMPS) (Genchi et al. 2020), monomethyl DMSA (Genchi et al. 2020), monoisoamyl DMSA (Genchi et al. 2020), and monocyclo hexyl DMSA (Genchi et al. 2020) is effective in acute Cd toxicity but not in chronic cases (Hoffman et al. 2023b). Additionally, N-acetyl cysteine, biologically active compounds, and antioxidants could be promising adjuvants (Genchi et al. 2020).

Cd exposure in mice enhances lipid peroxidation, glutamate oxaloacetate transaminase, and glutamate pyruvate transaminase activities. It decreases reduced glutathione (GSH) and serum alkaline phosphatase activity. Notably,

ginseng reverses these alterations (Shukla and Kumar 2009). A similar investigation indicated that Korean red ginseng protected against Cd-induced hepatotoxicity and contributed to the suppression of ALT, AST, and LDH in serum, amelioration of centrilobular necrosis around central veins, and the peripheral hemorrhage around portal triads. Moreover, Korean red ginseng restores the deficit in hepatic GSH content resulting from cadmium exposure (Park et al. 2013). A recent study proved the protective effects of Korean red ginseng against Cd lung toxicity and inflammation through inhibition of MAPK/extracellular signal-regulated kinase (ERK)1/2/activator protein 1 (AP-1) pathway (Mitra et al. 2022).

Shojaeepour et al. showed that *P. ginseng* extract against Cd-induced testicular toxicity in rats through the reduction of apoptosis of germ cells and a decrease in the MDA level. This extract ameliorates the toxic effects of Cd on testis's structure and sperm parameters and reverses the total antioxidant capacity (Shojaeepour et al. 2021).

Chrome

Hexavalent chromium [Cr (VI)] is a well-known carcinogen and toxic element on Earth that can lead to occupational toxicity through inhalation. The toxic effects of Cr (VI) compounds are associated with the production of ROS, the development of oxidative stress, and the disruption of mitochondria, DNA, RNA, and protein molecules (Singh et al. 2022). Acute ingestion of chromium can result in GI hemorrhage with or without bowel perforation. Acute dermal exposure can cause inflammation and ulceration. Other manifestations of acute chromium toxicity include coagulopathy, nephrotoxicity, and metabolic abnormalities. Chronic respiratory exposure can damage the respiratory system. Occupational exposure may lead to dermatitis toxicities and “blackjack disease.” Diagnostic tests for chromium toxicity include measuring chromium levels in blood and urine (Hoffman et al. 2023c).

It has been reported that Cr (VI) exposure in rats significantly increases ALT, AST, ALP, urea, and creatinine and decreases total protein, albumin, testosterone, GPx, and SOD in the liver, kidney, and testes tissue. At the same time, the ginseng treatment reverses all these changes and ameliorates the tissue damage (Abdel-Wahhab and Ahmed 2004). Elsaieed and Nada's investigation indicated that Cr (VI) treatment in female rats induces teratogenicity, and ginseng supplementation could reduce this side effect (Elsaieed and Nada 2002).

One study aimed to investigate the protective effects of ginseng polysaccharides against hexavalent chromium-induced kidney damage. The findings revealed that ginseng polysaccharides significantly lowered kidney damage markers such as kidney index, urea nitrogen, and serum creatinine

in treated mice. Additionally, these polysaccharides mitigated oxidative stress, apoptosis, and biofilm damage in renal tissues. A lipidomic analysis suggested that ginseng polysaccharides exert protection by modulating differential lipid expression (Jing et al. 2024).

Copper

Copper can cause acute toxicity, leading to gastrointestinal (GI) problems, especially oral toxicity, hemolysis, methemoglobinemia, hepatotoxicity, jaundice, kidney damage, lung dysfunction, hypotension, heart failure, and reduced mental function. Cough, chills, chest pain, and fever can occur from acute inhaling of copper. Long-term exposure to copper orally can result in cirrhosis. Chronic inhalation of copper can lead to interstitial pulmonary fibrosis, histiocytic granulomas with copper, lung adenocarcinoma, and hepatic angiosarcoma. Ophthalmic effects may also appear after continuous occupational exposure (Hoffman et al. 2023d).

At the cellular level, copper can cause oxidative stress, DNA damage, and cell injuries. To evaluate copper toxicity, tests for urine and blood copper levels, as well as serum ceruloplasmin levels, are necessary, in addition to assessing kidney and liver function. Hemolysis and decreased blood glutathione levels may also occur. Pharmacologic doses of zinc can reduce copper absorption in the GI by competing in absorption and inducing enterocyte metal chelator, metallothionein. Chelators such as EDTA, dimercaptopropane sulfonic acid, and especially D-penicillamine are commonly used in treating copper toxicity; but, D-penicillamine may be intolerable for 30% of patients, in which case trientine is recommended. In severe cases, a liver transplant may be necessary. Other treatment options may include ammonium tetrathiomolybdate, plasmapheresis, exchange transfusion, molecular adsorbent recirculating system, dialysis, and supportive care (Royer 2023).

One study investigated the protective effects of *P. ginseng* against liver damage caused by copper sulfate in rats. Results showed that copper sulfate significantly increased liver enzymes, copper density, and oxidative stress markers, indicating liver damage. However, rats treated with *P. ginseng* exhibited reduced inflammation and oxidative stress. Furthermore, *P. ginseng* enhanced liver glycogen density, glutathione levels, and superoxide dismutase activity, suggesting its potential to mitigate copper-induced liver toxicity (Yildiz et al. 2023).

Iron

Iron toxicity is one of the most dangerous toxicities in pediatrics, as it can lead to organ failure and, ultimately, death. The first signs of oral iron toxicity are gastrointestinal

symptoms, which can be followed by shock, metabolic acidosis, coagulopathy, hepatic dysfunction, cardiomyopathy, and renal failure. At the cellular level, excess free iron accumulates in the mitochondria, leading to oxidative stress, cell injury, and cell death. Between 2 and 8 weeks after iron toxicity, scarring and obstruction of the pyloric or proximal bowel may occur. Serum iron levels initially rise within 6 h of iron ingestion but decrease as the iron is cleared from the serum and deposited in the liver. Recommended tests include measuring electrolytes, kidney function, serum glucose, coagulation studies, complete blood count, liver function, and plain radiographs. IV crystalloid is infused to cope with hypovolemia and hypoperfusion. Deferoxamine is the primary chelator used to remove iron from tissue and plasma. Whole-bowel irrigation and gastric lavage are recommended to clear the GI system. Vitamin K and fresh frozen plasma are recommended in iron toxicity-induced coagulopathy (Yuen Hw 2023).

In human SK-N-SH, Rg-1 neuroblastoma cell line pre-exposure counteracts iron neurotoxicity. Rg-1 premedication increases cell viability and upregulates the HO-1 and Cu/Zn SOD; it induces Nrf2 nuclear translocation and activates PI3K/Akt pathway (Du et al. 2013). The PI3K/Akt pathway has a fundamental role in neuronal survival. Akt is one of the downstream targets of PI3K (Chong et al. 2005), and Nrf2 is upstream of antioxidant enzyme HO-1 expression (Leiser and Miller 2010).

Lead (Pb)

Pb is a naturally toxic metal and is a major cause of occupational toxicity in the industry (Nava-Ruiz et al. 2012). Plumbism can lead to cognitive and neurobehavioral dysfunctions, hypertension, and neurological and renal impairments (Patrick 2006). Lead interferes with the activity of divalent cations, such as calcium, and disturbs their roles in cellular mechanisms. The toxic effects of lead can include neurological problems such as seizures and coma, as well as anemia, Fanconi-like syndrome, saturnine gout, hypertension, and cardiovascular disease. Lead toxicity is also associated with disrupting GI and thyroid function, affecting growth, skeletal development, and reproductivity (Halmo 2023). Laboratory evaluation tests can include measuring blood lead concentration, erythrocyte protoporphyrin concentration, kidney function tests, liver function tests, urinalysis, and complete blood count. Radiographic studies can help identify the presence of foreign lead bodies and lead lines (Hoffman et al. 2023e).

To eliminate lead from the body, treatments such as laxatives, bowel irrigation, and surgery may be useful. Chelating agents like succimer, calcium disodium ethylenediaminetetraacetic acid (EDTA), and British anti-Lewisite (BAL, also known as dimercaprol) may also be used, although they may

Table 2 Summary of the protective effects of *Panax ginseng* and its main components against the deteriorative effects of metals

Toxic agent	Dose, duration, and route of exposure to toxicant	Ginseng derivatives, dose, duration, route of exposure	Experimental model	Results	References
Aluminum	D-Gal (150 mg/kg), s.c., with AlCl ₃ (13 mg/kg) intragastrically, once a day for 42 days	Rg1 (10 and 20 mg/kg/day), i.p., for 20 days, at the end of exposure with toxicant	Induced aging in male Kunming mice	↑ Spatial learning and memory ↓ Apoptosis in the hippocampus and prefrontal cortex ↑ FGF2, BDNF, and TrkB/Akt signaling in the hippocampus and prefrontal cortex	(Zhong et al. 2020)
	D-Gal (60 mg/kg), i.p. with AlCl ₃ (40 mg/kg) intragastrically, once a day for 90 days	Ginseng protein (0.1 g/kg), BID, from day 60 for 30 days, intragastrically	Induced Alzheimer's disease in female Wistar rats	↑ The survival rate of neurons ↑ PI3K, p Akt/Ak ↓ Cells' apoptosis ↓ Caspase-3 and Bax/Bcl-2 - Improved cognition ↓ Aβ₁₋₄₂ and p-tau	(Li et al. 2016)
Al	(200 mg/kg) intragastrical and D-gal (60 mg/kg) i.p. for 60 days	From day 30, the ginsenosides (2 g crude drug/kg/day) for 30 days, p.o	Induced special memory impairment in male Wistar rats	↑ Cognitive behavioral ↓ Expressions of Aβ and p-tau ↑ γ-Aminobutyric acid, acetylcholine, and dopamine levels in the hippocampus and cortex ↓ Glutamate and aspartic acid levels in the hippocampus and cortex ↑ Glycine and serotonin levels in the blood	(Zhang et al. 2016)
	AlCl ₃ (64 mg/kg/day) p.o. for 120 days	Rg3 (20 mg/kg/day), from the 91st, mf p.o. for 30 days	AlCl ₃ -induced osteoporosis in male Sprague–Dawley rats	↑ Bone mineral density ↓ ROS and MDA ↑ GPx and SOD activities ↑ Concentration of Ca²⁺ ↓ Content of N-terminal cross-linking telopeptide of type I collagen, C-terminal cross-linking telopeptide of type I collagen, and tartrate-resistant acid phosphatase 5b activity in serum ↑ TGF-β1, BMP-2, IGF-I, and Cbfa1	(Song et al. 2020)
	AlCl ₃ ·6H ₂ O (0.126 mg/mL) for 24 h	Rb1 0.0145 mg/mL for 24 h, concomitant with AlCl ₃	AlCl ₃ -induced osteoblasts dysfunction, osteoblasts isolated from 3-day-old newborn Wistar rats	↑ Cell viability, TGF-β1, BMP-2, IGF-I, and Cbfa1 mRNA expressions ↑ GPX and SOD activities ↓ ROS concentration and osteoblasts histology lesions	(Zhu et al. 2016)
	AlCl ₃ 200 mg/kg/day for 6 months, in drinking water	Post-treatment with Rb1 (20 mg/kg/day) for another 4 months, in drinking water	AlCl ₃ -induced cognitive deficits in female imprinting control region (ICR) mice	↑ Cognitive behavioral ↓ tau phosphorylation in the brain ↓ p-GSK3in the brain ↑ PP2A in the brain	(Zhao et al. 2013)

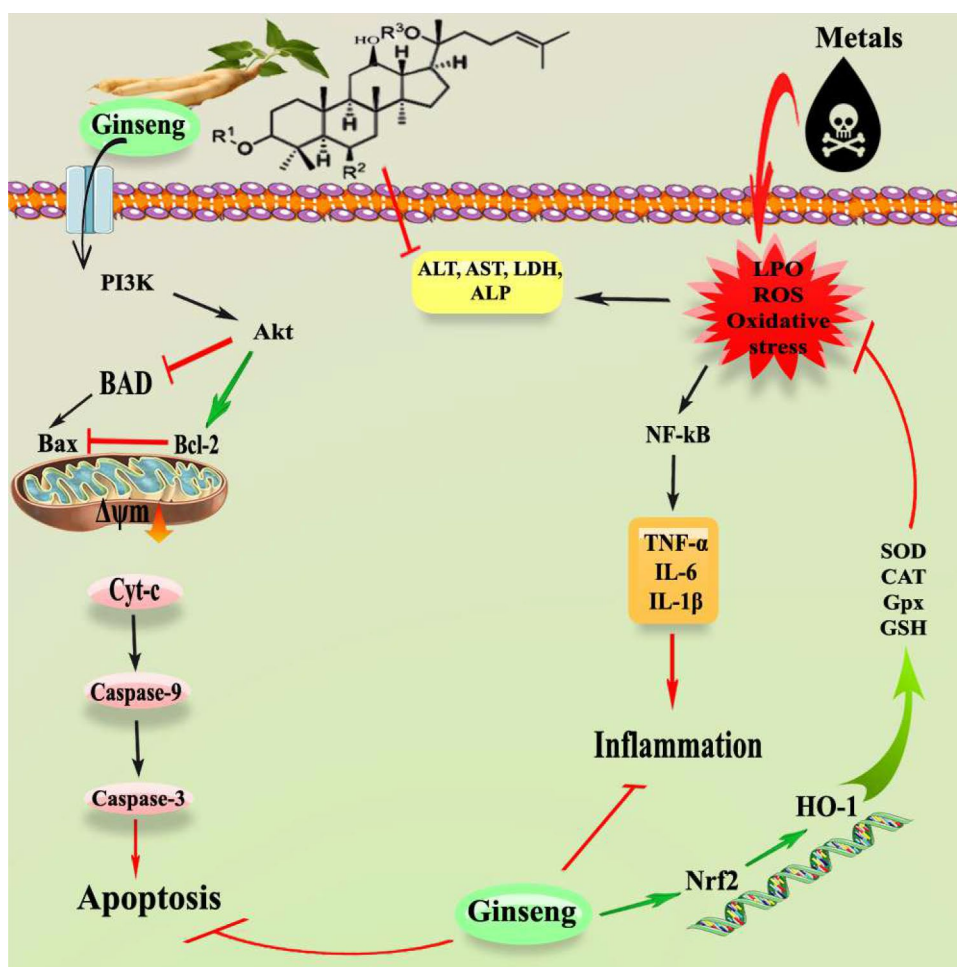
Table 2 (continued)

Toxic agent	Dose, duration, and route of exposure to toxicant	Ginseng derivatives, dose, duration, route of exposure	Experimental model	Results	References
Cadmium	CdCl ₂ (1 mg/kg) i.p. at once	Ginseng root extract (10 mg/kg), for 10 days prior to exposure to cadmium, p.o	Cadmium-induced hepatic injuries in male Swiss albino mice	↓ LPO, SGOT and SGPT ↑ GSH and ALP	(Shukla and Kumar 2009)
	CdCl ₂ (4 mg/kg) single dose, i.v. after the last dose of KRG	KRG (50 mg/kg) for 7 days, p.o	Cadmium-induced-hepatic injuries in Sprague–Dawley rats	↓ ALT, AST, and LDH Improving centrilobular necrosis around central veins and the peripheral hemorrhage ↑ GSH ↓ Mitochondrial Bad ¹	(Park et al. 2013)
	1.2 mg/kg single dose on the first day, i.p	<i>P. ginseng</i> extract (50 mg/kg) for 9 weeks after Cd injection, p.o	Cadmium-induced spermatogenesis damage in male Wistar rats	↓ Apoptosis of germ cell and of MDA levels ↑ Testis structure and sperm parameters	(Shojaeepour et al. 2021)
Chromium	Cr VI (50 mg/kg) for 15 days, p.o	KRG (20 mg/kg) for 15 days (before or concomitant or after), p.o	Chromium-induced liver and kidney damages in male Sprague–Dawley rats	↓ ALT, AST, ALP, urea, and creatinine ↑ Albumin, testosterone, GPx and SOD	(Abdel-Wahhab and Ahmed 2004)
	Hexavalent chromium (50 mg/kg); p.o	Ginseng polysaccharide (25, 50, 100, 200, and 400 mg/kg)	Hexavalent chromium-induced nephrotoxicity in mice	↓ Kidney injury index (ipsilateral kidney mass/body weight) ↓ Urea nitrogen ↓ Serum creatinine ↓ Oxidative stress ↓ Apoptosis	(Jing et al. 2024)
Iron	Fe ²⁺ (100 μM) for 4 h	Rg1 (0.1, 1 μM) 24 h before exposure to Iron	Iron-induced neurotoxicity in human SK-N-SH neuroblastoma cells	↑ Cell viability PI3K/Akt, Nrf2, HO-1, and Cu/Zn-SOD	(Du et al. 2013)
Lead	0.3% lead acetate in drinking water from 2 weeks before pregnancy to postnatal day 21 Water containing lead acetate (0.2%) for 12 weeks, before exposure to Rg1	Gintonin (100 mg/kg), from 2 weeks before pregnancy to postnatal day 21, p.o Rg1 (40 mg/kg) for 4 weeks, i.p. post-exposure to toxicant	Lead-induced cerebellar impairments in postnatal Sprague–Dawley rats Lead-induced hematopoietic homeostasis defects in Sprague–Dawley rats	↓ Bax, oxidative stress, IL-1β, and TNF-α ↑ Bcl-2, Nrf2 and SOD ↓ Inflammation (TNF-α and IL-6) ↓ Apoptosis Improve the life span of erythrocytes ↑ CD4 ⁺ and CD8 ⁺ T lymphocytes	(Nam et al. 2020) (Wang et al. 2018)
Copper	100 mg/kg/day (CuSO ₄ , Anhydrous) gavage, for 14 days	<i>P. ginseng</i> (300 mg/kg/day), gavage for 14 days	Copper sulfate-induced liver toxicity in rats	↓ Inflammatory cell infiltration in liver ↓ Hepatic MDA level ↓ Hepatic glycogen density ↑ Hepatic GSH level ↑ hepatic SOD level	(Yildiz et al. 2023)

Akt protein kinase B, *ALP* alkaline phosphatase, *Al* aluminum, *ALT* alanine aminotransferase, *AST* aspartate aminotransferase, *Bax* Bcl-2-associated X protein, *Bcl-2* B-cell lymphoma 2, *BDNF* brain-derived neurotrophic factor, *BID* twice a day, *BMP-2* bone morphogenetic protein-2, *cAMP* cyclic adenosine monophosphate, *Chfr1* core-binding factor α1, *D-Gal* D-galactose, *FGF2* fibroblast growth factor 2, *GPX* glutathione peroxidase, *GSH* glutathione, *GSK3* glycogen synthase kinase 3, *HO-1* heme oxygenase-1, *IGF-1* insulin-like growth factor 1, *i.p.* intraperitoneal, *iNOS* inducible nitric oxide synthase, *IL-* interleukin-, *i.v.* intravenous, *KRG* Korean red ginseng, *LDH* lactate dehydrogenase, *LPO* lipid peroxidation, *MDA* malondialdehyde, *Nrf2* nuclear erythroid-2-related factor2, *PI3K* phosphatidylinositol-3-kinase, *p.o.* per oral, *PP2A* protein phosphatase-2A, *ROS* reactive oxygen species, *SGOT* serum glutamate oxaloacetate transaminase, *SGPT* serum glutamate pyruvate transaminase, *SOD* superoxide dismutase, *TGF-β1* transforming growth factor-β1, *t-NOS* total nitric oxide synthase, *TrkB* tropomyosin receptor kinase B

¹ Mitochondrial Bad protein: a representative apoptosis marker

Fig. 5 The molecular and cellular protective mechanisms of ginseng and its main constituents against metal toxicity. Akt, protein kinase B; ALT, alanine transaminase; AST, aspartate aminotransferase; Bax, Bcl-2-associated X protein; Bcl-2, B-cell lymphoma 2; CAT, catalase; cyt-c, cytochrome-c; GPx, glutathione peroxidase; GSH, glutathione; HO-1, heme oxygenase; IL-, interleukin; LDH, lactate dehydrogenase; MAPK, mitogen-activated protein kinase; mTOR, the mammalian target of rapamycin; NF- κ B, nuclear factor κ -light-chain enhancer of activated B cells; Nrf2, nuclear factor erythroid 2-related factor 2; PI3K, phosphatidylinositol 3-kinase; ROS, reactive oxygen species; SOD, superoxide dismutase; TNF- α , tumor necrosis factor- α



potentially cause acute lead toxicity by mobilizing lead from stable compartments to unstable ones (Halmo 2023).

It has been reported that gintonin administration attenuates Pb-induced damage in the developing cerebellum and improves locomotive impairment in pups from mothers exposed to Pb during pregnancy and lactation (Nam et al. 2020).

Acute Pb exposure could further attenuate neurogenesis in the brains of aged rats in compare to the average age. It stimulates microgliosis and inflammation and inhibits the proliferation of neural progenitors. Ginsenoside Rd3 meaningfully reversed these deteriorative effects (Wang et al. 2013). In addition, Rg1 remarkably ameliorates the lead acetate-mediated hematopoietic stem cells (HSCs) aging by alleviating the senescence-associated p53-p21-Rb signaling network in the hematopoietic system (Wang et al. 2018).

Based on previous research projects, ginseng and its constituents have antidote/protective effects against metal toxicity (Table 2) and reduce the teratogenicity and detrimental effects of metals on the testis, bones, liver, lung, and neuronal system. In these reviewed studies, no study mentions their chelating properties, but most of them mainly induce

these protective effects via activating anti-inflammatory, antioxidant, and anti-apoptotic pathways. Some restore the effect of BDNF and FGFs on PI3K/Akt and Bcl-2 signaling pathways and inhibit caspase-3 activation; some trigger the cAMP/PKA/CREB signaling pathway and increase the expression of osteoblast growth regulatory factors. Inhibition of the MAPK/ERK1/2/AP-1 pathway is another suggested anti-inflammatory pathway for some ginseng derivatives. Furthermore, some of these compounds diminish tau phosphorylation by decreasing p-GSK-3 and increasing the PP2A level. (Fig. 5).

Conclusion and future direction

Natural toxins are poisonous substances produced by living organisms with no harmful effects on themselves. They are categorized as fungal, microbial, herbal, algae, and animal-based toxins. From one viewpoint, metals are also naturally occurring toxins on Earth. They are not created or destroyed, but their concentration in the biosphere has simply changed.

Several investigations indicate the protective effects of ginseng and its main secondary metabolites against natural toxins and metal toxicity. They have anti-oxidative, anti-inflammatory, and anti-apoptotic effects through different cell signaling pathways.

By activating the Nrf2/HO-1 signaling pathway and increasing the levels of SOD, CAT, GPX, and GSH, they play a key role in reducing ROS and oxidative stress. By blocking the PERK/eIF2 α /ATF4 signaling pathway and decreasing the levels of Bax/Bcl-2 and caspase-3, caspase-8, and caspase-9, they inhibit apoptosis. Additionally, by downregulating TLR-2 expression, inhibiting the MAPK/ERK1/2/AP-1 pathway, and decreasing iNOS, TNF- α , IL-6, and IL-1 β , they exhibit anti-inflammatory effects. They also restore the effect of BDNF and FGFs on the PI3K/Akt pathway (Figs. 4 and 5).

The results were gathered from in vitro and in vivo studies, and no related clinical trials were found to be included; limited data in human subjects is a negative point. This may be related to limited and unpredicted human exposure to toxins and ethical codes that restrict research in this field. However, to apply the findings of the basic sciences to clinical practice, results from clinical studies are crucial.

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Author contribution HH was responsible for conceptualization, writing –review & editing, supervision, methodology, and data curation. MR, ZM, and KN were responsible for writing –the original draft, review & editing, methodology– literature search, and data curation. All the authors read and approved the final version of the manuscript. The authors confirm that no paper mill and artificial intelligence were used. All authors agree to be accountable for all aspects of work ensuring integrity and accuracy. During the preparation of this work, the author(s) used AI-assisted technologies to rephrase to reduce plagiarism and improve the language and grammar. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Data availability No datasets were generated or analysed during the current study.

Declarations

Human ethics and consent to participate Not applicable.

Competing interests The authors declare no competing interests and also H.H. is the Editor of Naunyn-Schmiedeberg's Arch Pharmacol and followed the submission guidelines of the journal (https://link.springer.com/journal/210/submission-guidelines#Instructions%20for%20Authors_Competing%20Interests).

References

Abdel-Aziem SH, Hassan AM, Abdel-Wahhab MA (2011) Dietary supplementation with whey protein and ginseng extract counteracts oxidative stress and DNA damage in rats fed an

aflatoxin-contaminated diet. *Mutat Res* 723(1):65–71. <https://doi.org/10.1016/j.mrgentox.2011.04.007>

Abdel-Wahhab MA, Ahmed HH (2004) Protective effect of Korean Panax ginseng against chromium VI toxicity and free radicals generation in rats. *J Ginseng Res* 28(1):11–17

Abdel-Wahhab MA, Hassan NS, El-Kady AA, Khadrawy YA, El-Nekeety AA, Mohamed SR, Sharaf HA, Mannaa FA (2010) Red ginseng extract protects against aflatoxin B1 and fumonisins-induced hepatic pre-cancerous lesions in rats. *Food Chem Toxicol* 48(2):733–42. <https://doi.org/10.1016/j.fct.2009.12.006>

Abdelghany AK, Khalil F, Azeem NMA, El-Nahass ES, El-Kashlan AM, Emeash HH (2019) Ginseng and moringa olifera ameliorated cognitive impairments induced by aluminium chloride in albino rat. *Adv Anim Vet Sci* 7(7):557–565. <https://doi.org/10.17582/journal.aavs/2019/7.7.557.565>

Abdulateef SM, Ibraheem SR, Hussein HS, Dheeb BI, Khashman BM, Ahmed DM, Abu-Elteen KH, Abu-Qatouseh L (2024) MMP-1 and MMP-7 expression is influenced by ginsenosides in mice exposed to aflatoxin B1: in vivo study. *Jordan J Pharm Sci* 17(1):199–214. <https://doi.org/10.35516/jjps.v17i2.1989>

Aborehab NM, Waly NE (2019) IL-6 and NFE2L2: a putative role for the hepatoprotective effect of N. Sativa, P. Ginseng and C. Sempervirens in AFB-1 induced hepatocellular carcinoma in rats. *Toxicol Rep* 6:457–464. <https://doi.org/10.1016/j.toxrep.2019.05.008>

Ahmadi A, Mortazavi Z, Mehri S, Hosseinzadeh H (2022) Scutellaria baicalensis and its constituents baicalin and baicalein as antidotes or protective agents against chemical toxicities: a comprehensive review. *Naunyn Schmiedeberg's Arch Pharmacol* 395(11):1297–1329. <https://doi.org/10.1007/s00210-022-02258-8>

Al-Mudallal N (2023) The expression of MMP1 and MMP7 in mice liver after exposure to aflatoxin B1 using immunohistochemistry technique. *Arch Razi Inst* 78(1): 63–72. <https://doi.org/10.22092/ARI.2022.358774.2306>

Alavi MS, Fanoudi S, Ghasemzadeh Rahbardi M, Mehri S, Hosseinzadeh H (2021) An updated review of protective effects of rosemary and its active constituents against natural and chemical toxicities. *Phytother Res* 35(3):1313–1328. <https://doi.org/10.1002/ptr.6894>

Aminifard T, Razavi BM, Hosseinzadeh H (2021) The effects of ginseng on the metabolic syndrome: an updated review. *Food Sci Nutr* 9(9):5293–5311. <https://doi.org/10.1002/fsn3.2475>

Ang-Lee MK, Moss J, Yuan CS (2001) Herbal medicines and perioperative care. *JAMA* 286(2):208–16. <https://doi.org/10.1001/jama.286.2.208>

Ardah MT, Paleologou KE, Lv G, Menon SA, Abul Khair SB, Lu JH, Safieh-Garabedian B, Al-Hayani AA et al (2015) Ginsenoside Rb1 inhibits fibrillation and toxicity of alpha-synuclein and dis-aggregates preformed fibrils. *Neurobiol Dis* 74:89–101. <https://doi.org/10.1016/j.nbd.2014.11.007>

Bak D-H, Kim HD, Kim YO, Park CG, Han S-Y, Kim J-J (2016) Neuroprotective effects of 20(S)-protopanaxadiol against glutamate-induced mitochondrial dysfunction in PC12 cells. *Int J Mol Med* 37(2):378–386. <https://doi.org/10.3892/ijmm.2015.2440>

Bertin G, Averbeck D (2006) Cadmium: cellular effects, modifications of biomolecules, modulation of DNA repair and genotoxic consequences (a review). *Biochimie* 88(11):1549–59. <https://doi.org/10.1016/j.biochi.2006.10.001>

Cagnol S, Chambard JC (2010) ERK and cell death: mechanisms of ERK-induced cell death–apoptosis, autophagy and senescence. *FEBS J* 277(1):2–21. <https://doi.org/10.1111/j.1742-4658.2009.07366.x>

Cataldi M (2010) Batrachotoxin. In: Enna SJ, Bylund DB (eds.) *xPharm: the comprehensive pharmacology reference*. Elsevier, New York

- Catterall WA, Cestèle S, Yarov-Yarovoy V, Yu FH, Konoki K, Scheuer T (2007) Voltage-gated ion channels and gating modifier toxins. *Toxicol* 49(2):124–41. <https://doi.org/10.1016/j.toxicol.2006.09.022>
- Chacko B, Peter JV (2019) Antidotes in poisoning. *Indian J Crit Care Med* 23(Suppl 4):S241–s249. <https://doi.org/10.5005/jp-journals-10071-23310>
- Choi S-H, Jung S-W, Lee B-H, Kim H-J, Hwang S-H, Kim H-K, Nah S-Y (2015) Ginseng pharmacology: a new paradigm based on gintonin-lysophosphatidic acid receptor interactions. *Front Pharmacol* 6(245). <https://doi.org/10.3389/fphar.2015.00245>
- Choi S, Lim JW, Kim H (2020) Korean red ginseng inhibits amyloid- β -induced apoptosis and Nuclein expression in human neuronal cells. *Pharmacology* 105(9–10):584–595
- Chong ZZ, Li F, Maiese K (2005) Oxidative stress in the brain: novel cellular targets that govern survival during neurodegenerative disease. *Prog Neurobiol* 75(3):207–46. <https://doi.org/10.1016/j.pneurobio.2005.02.004>
- Cope WG, Hodgson E (2010) Classes of toxicants: use classes. In: Hodgson E (ed.) *A textbook of modern toxicology*. 4 ed. John Wiley & Sons, New Jersey
- Cui J, Shan R, Cao Y, Zhou Y, Liu C, Fan Y (2021) Protective effects of ginsenoside Rg2 against memory impairment and neuronal death induced by A β 25–35 in rats. *J Ethnopharmacol* 266:113466
- De Punder K, Pruimboom L (2015) Stress induces endotoxemia and low-grade inflammation by increasing barrier permeability. *Front Immunol* 6:223. <https://doi.org/10.3389/fimmu.2015.00223>
- Dorri M, Hashemitabar S, Hosseinzadeh H (2018) Cinnamon (*Cinnamomum zeylanicum*) as an antidote or a protective agent against natural or chemical toxicities: a review. *Drug Chem Toxicol* 41(3):338–351. <https://doi.org/10.1080/01480545.2017.1417995>
- Dorrigiv M, Zareiyan A, Hosseinzadeh H (2020) Garlic (*Allium sativum*) as an antidote or a protective agent against natural or chemical toxicities: a comprehensive update review. *Phytother Res* 34(8):1770–1797. <https://doi.org/10.1002/ptr.6645>
- Dorrigiv M, Zareiyan A, Hosseinzadeh H (2021) Onion (*Allium cepa*) and its main constituents as antidotes or protective agents against natural or chemical toxicities: a comprehensive review. *Iran J Pharm Res* 20(1):3–26. <https://doi.org/10.22037/ijpr.2020.112773.13940>
- Du X, Xu H, Jiang H, Xie J (2013) Akt/Nrf2 activated upregulation of heme oxygenase-1 involves in the role of Rg1 against ferrous iron-induced neurotoxicity in SK-N-SH cells. *Neurotox Res* 24(1):71–79. <https://doi.org/10.1007/s12640-012-9362-3>
- Duan Y, Zheng J, Law V, Nicholson R (2006) Natural products from ginseng inhibit [3H]batrachotoxinin A 20- α -benzoate binding to Na⁺ channels in mammalian brain. *Eur J Pharmacol* 530(1–2):9–14. <https://doi.org/10.1016/j.ejphar.2005.11.022>
- El-Demerdash FM, El-Magd MA, El-Sayed RA (2021) Panax ginseng modulates oxidative stress, DNA damage, apoptosis, and inflammations induced by silicon dioxide nanoparticles in rats. *Environ Toxicol* 36(7):1362–1374. <https://doi.org/10.1002/tox.23132>
- Elsaieed E, Nada S (2002) Teratogenicity of hexavalent chromium in rats and the beneficial role of ginseng. *Bull Environ Contam Toxicol* 68(3):361–368
- Fanoudi S, Alavi MS, Karimi G, Hosseinzadeh H (2020) Milk thistle (*Silybum Marianum*) as an antidote or a protective agent against natural or chemical toxicities: a review. *Drug Chem Toxicol* 43(3):240–254. <https://doi.org/10.1080/01480545.2018.1485687>
- Forde JE, Dale TC (2007) Glycogen synthase kinase 3: a key regulator of cellular fate. *Cell Mol Life Sci* 64(15):1930–1944. <https://doi.org/10.1007/s00018-007-7045-7>
- Gao Y, Chu SF, Li JP, Zhang Z, Yan JQ, Wen ZL, Xia CY, Mou Z et al (2015) Protopanaxatriol protects against 3-nitropropionic acid-induced oxidative stress in a rat model of Huntington's disease. *Acta Pharmacol Sin* 36(3):311–322. <https://doi.org/10.1038/aps.2014.107>
- Genchi G, Sinicropi M S, Lauria G, Carocci A, Catalano A (2020) The effects of cadmium toxicity. *Int J Environ Res Public Health* 17(11). <https://doi.org/10.3390/ijerph17113782>
- Ghorbani Z, Mirghafourvand M, Charandabi SM-A, Javadzadeh Y (2019) The effect of ginseng on sexual dysfunction in menopausal women: a double-blind, randomized, controlled trial. *Complement Ther Med* 45:57–64. <https://doi.org/10.1016/j.ctim.2019.05.015>
- Gu Y, Ren K, Wang L, Jiang C, Yao Q (2020) Rg1 in combination with mannitol protects neurons against glutamate-induced ER stress via the PERK-eIF2 α -ATF4 signaling pathway. *Life Sci* 263:118559
- Haddad JJ (2005) N-methyl-D-aspartate (NMDA) and the regulation of mitogen-activated protein kinase (MAPK) signaling pathways: a revolving neurochemical axis for therapeutic intervention? *Prog Neurobiol* 77(4):252–82. <https://doi.org/10.1016/j.pneurobio.2005.10.008>
- Halmo LNT (2023) Lead toxicity. StatPearls [Internet]. StatPearls Publishing, Treasure Island (FL)
- Hanger DP, Hughes K, Woodgett JR, Brion JP, Anderton BH (1992) Glycogen synthase kinase-3 induces Alzheimer's disease-like phosphorylation of tau: generation of paired helical filament epitopes and neuronal localisation of the kinase. *Neurosci Lett* 147(1):58–62. [https://doi.org/10.1016/0304-3940\(92\)90774-2](https://doi.org/10.1016/0304-3940(92)90774-2)
- Hassan AM, Abdel-Aziem SH, El-Nekeety AA, Abdel-Wahhab MA (2015) Panax ginseng extract modulates oxidative stress, DNA fragmentation and up-regulate gene expression in rats sub chronically treated with aflatoxin B1 and fumonisin B 1. *Cytotechnology* 67(5):861–71. <https://doi.org/10.1007/s10616-014-9726-z>
- He L-X, Zhang Z-F, Zhao J, Li L, Xu T, Bin S, Ren J-W, Liu R et al (2018) Ginseng oligopeptides protect against irradiation-induced immune dysfunction and intestinal injury. *Sci Rep* 8(1):13916. <https://doi.org/10.1038/s41598-018-32188-6>
- He X, Li Y, Deng B, Lin A, Zhang G, Ma M, Wang Y, Yang Y et al (2022) The PI3K/AKT signalling pathway in inflammation, cell death and glial scar formation after traumatic spinal cord injury: mechanisms and therapeutic opportunities. *Cell Prolif* 55(9):e13275. <https://doi.org/10.1111/cpr.13275>
- Hedayati N, Naeini MB, Nezami A, Hosseinzadeh H, Wallace Hayes A, Hosseini S, Imenshahidi M, Karimi G (2019) Protective effect of lycopene against chemical and natural toxins: a review. *BioFactors* 45(1):5–23. <https://doi.org/10.1002/biof.1458>
- Heidari S, Mehri S, Hosseinzadeh H (2021) The genus *Glycyrrhiza* (Fabaceae family) and its active constituents as protective agents against natural or chemical toxicities. *Phytother Res* 35(12):6552–6571. <https://doi.org/10.1002/ptr.7238>
- Hoffman RS, Gosselin S, Nelson LS, Howland MA, Lewin NA, Smith SW, Goldfrank LR (2023a) Aluminum. In: Goldfrank's clinical manual of toxicologic emergencies, 2nd edn. McGraw Hill, New York, pp 445–447
- Hoffman RS, Gosselin S, Nelson LS, Howland MA, Lewin NA, Smith SW, Goldfrank LR (2023b) Cadmium. In: Goldfrank's clinical manual of toxicologic emergencies, 2nd edn. McGraw Hill, New York, pp 461–462
- Hoffman RS, Gosselin S, Nelson LS, Howland MA, Lewin NA, Smith SW, Goldfrank LR (2023c) Chromium. In: Goldfrank's clinical manual of toxicologic emergencies, 2nd edn. McGraw Hill, New York, pp 465–467
- Hoffman RS, Gosselin S, Nelson LS, Howland MA, Lewin NA, Smith SW, Goldfrank LR (2023d) Copper. In: Goldfrank's clinical manual of toxicologic emergencies, 2nd edn. McGraw Hill, New York, pp 471–473
- Hoffman RS, Gosselin S, Nelson LS, Howland MA, Lewin NA, Smith SW, Goldfrank LR (2023e) Lead. In: Goldfrank's clinical manual

- of toxicologic emergencies, 2nd edn. McGraw Hill, New York, pp 474–489
- Hosseini A, Hosseinzadeh H (2018) Antidotal or protective effects of *Curcuma longa* (turmeric) and its active ingredient, curcumin, against natural and chemical toxicities: a review. *Biomed Pharmacother* 99:411–421. <https://doi.org/10.1016/j.biopha.2018.01.072>
- Hosseini A, Razavi BM, Hosseinzadeh H (2023) Protective effects of pomegranate (*Punica granatum*) and its main components against natural and chemical toxic agents: a comprehensive review. *Phytomedicine* 109:154581. <https://doi.org/10.1016/j.phymed.2022.154581>
- Hu Y, Xiang C, Zhang D, Zhou F, Zhang D (2024) Nephroprotective effect of Ginsenoside Rg1 in lipopolysaccharide-induced sepsis in mice through the SIRT1/NF- κ B signaling. *Folia Histochem Cytobiol* 62(1):13–24. <https://doi.org/10.5603/fhc.97140>
- Hynd MR, Scott HL, Dodd PR (2004) Glutamate-mediated excitotoxicity and neurodegeneration in Alzheimer's disease. *Neurochem Int* 45(5):583–595. <https://doi.org/10.1016/j.neuint.2004.03.007>
- Igbokwe IO, Igwenagu E, Igbokwe NA (2019) Aluminium toxicosis: a review of toxic actions and effects. *Interdiscip Toxicol* 12(2):45–70. <https://doi.org/10.2478/intox-2019-0007>
- Imai H, Nakagawa Y (2003) Biological significance of phospholipid hydroperoxide glutathione peroxidase (PHGPx, GPx4) in mammalian cells. *Free Radic Biol Med* 34(2):145–69. [https://doi.org/10.1016/s0891-5849\(02\)01197-8](https://doi.org/10.1016/s0891-5849(02)01197-8)
- Jang M, Choi JH, Chang Y, Lee SJ, Nah SY, Cho IH (2019) Gintonin, a ginseng-derived ingredient, as a novel therapeutic strategy for Huntington's disease: activation of the Nrf2 pathway through lysophosphatidic acid receptors. *Brain Behav Immun* 80:146–162. <https://doi.org/10.1016/j.bbi.2019.03.001>
- Jeffery B, Barlow T, Moizer K, Paul S, Boyle C (2004) Amnesic shellfish poison. *Food Chem Toxicol* 42(4):545–557. <https://doi.org/10.1016/j.fct.2003.11.010>
- Ji P, Shi Q, Kong L, Liu Y, Su Y, Sun R, Zhou H, Xu H et al (2024) Ginsenoside Rg1 attenuates chronic inflammation-induced renal fibrosis in mice by inhibiting AIM2 inflammasome in an Nrf2-dependent manner. *J Funct Foods* 116. <https://doi.org/10.1016/j.jff.2024.106204>
- Jing B, Wei M, Chen H, Xie W, An S, Li J, Wang S, Zhou X (2024) Pharmacodynamic evaluation and mechanism of ginseng polysaccharide against nephrotoxicity induced by hexavalent chromium. *Nutrients* 16(10). <https://doi.org/10.3390/nu16101416>
- Jung DH, Nahar J, Mathiyalagan R, Rupa EJ, Ramadhania ZM, Han Y, Yang DC, Kang SC (2023) A focused review on molecular signalling mechanisms of ginsenosides anti-lung cancer and anti-inflammatory activities. *Anticancer Agents Med Chem* 23(1):3–14. <https://doi.org/10.2174/1871520622666220321091022>
- Khandelwal PJ, Herman AM, Moussa CE (2011) Inflammation in the early stages of neurodegenerative pathology. *J Neuroimmunol* 238(1–2):1–11. <https://doi.org/10.1016/j.jneuroim.2011.07.002>
- Khoi CS, Chen JH, Lin TY, Chiang CK, Hung KY (2021) Ochratoxin A-induced nephrotoxicity: up-to-date evidence. *Int J Mol Sci* 22(20). <https://doi.org/10.3390/ijms222011237>
- Kim DH (2012) Chemical diversity of *Panax ginseng*, *Panax quinquefolium*, and *Panax notoginseng*. *J Ginseng Res* 36(1):1–15. <https://doi.org/10.5142/jgr.2012.36.1.1>
- Kim JH, Kim S, Yoon IS, Lee JH, Jang BJ, Jeong SM, Lee JH, Lee BH et al (2005) Protective effects of ginseng saponins on 3-nitropropionic acid-induced striatal degeneration in rats. *Neuropharmacology* 48(5):743–756. <https://doi.org/10.1016/j.neuropharm.2004.12.013>
- Kim YS, Kim YH, Noh JR, Cho ES, Park JH, Son HY (2011) Protective effect of Korean red ginseng against aflatoxin B1-induced hepatotoxicity in rat. *J Ginseng Res* 35(2):243–9. <https://doi.org/10.5142/jgr.2011.35.2.243>
- Kong L, Liu Y, Li J, Wang Y, Ji P, Shi Q, Han M, Xu H et al (2024) Ginsenoside Rg1 alleviates chronic inflammation-induced neuronal ferroptosis and cognitive impairments via regulation of AIM2 - Nrf2 signaling pathway. *J Ethnopharmacol* 330. <https://doi.org/10.1016/j.jep.2024.118205>
- Lai JL, Liu YH, Liu C, Qi MP, Liu RN, Zhu XF, Zhou QG, Chen YY et al (2017) Indirubin inhibits LPS-induced inflammation via TLR4 abrogation mediated by the NF- κ B and MAPK signaling pathways. *Inflammation* 40(1):1–12. <https://doi.org/10.1007/s10753-016-0447-7>
- Lee JH, Lee BH, Choi SH, Yoon IS, Shin TJ, Pyo MK, Lee SM, Kim HC et al (2008) Involvement of batrachotoxin binding sites in ginsenoside-mediated voltage-gated Na⁺ channel regulation. *Brain Res* 1203:61–67. <https://doi.org/10.1016/j.brainres.2008.01.078>
- Lee S, Rhee DK (2017) Effects of ginseng on stress-related depression, anxiety, and the hypothalamic-pituitary-adrenal axis. *J Ginseng Res* 41(4):589–594. <https://doi.org/10.1016/j.jgr.2017.01.010>
- Lee S, Youn K, Jun M (2018) Major compounds of red ginseng oil attenuate A β 25–35-induced neuronal apoptosis and inflammation by modulating MAPK/NF- κ B pathway. *Food Funct* 9(8):4122–4134
- Lee WJ, Kim EN, Trang NM, Lee JH, Cho SH, Choi HJ, Song GY, Jeong GS (2023) Ameliorative effect of ginsenoside Rg6 in periodontal tissue inflammation and recovering damaged alveolar bone. *Molecules* 29(1). <https://doi.org/10.3390/molecules29010046>
- Lee YY, Kim SD, Park JK, Lee WJ, Han JE, Seo MS, Seo MG, Bae S et al (2024) Red ginseng extract inhibits lipopolysaccharide-induced platelet-leukocyte aggregates in mice. *J Ginseng Res*. <https://doi.org/10.1016/j.jgr.2024.03.009>
- Leiser SF, Miller RA (2010) Nrf2 signaling, a mechanism for cellular stress resistance in long-lived mice. *Mol Cell Biol* 30(3):871–884. <https://doi.org/10.1128/mcb.01145-09>
- Li C, Gao H, Feng X, Bi C, Zhang J, Yin J (2020) Ginsenoside Rh2 impedes proliferation and migration and induces apoptosis by regulating NF- κ B, MAPK, and PI3K/Akt/mTOR signaling pathways in osteosarcoma cells. *J Biochem Mol Toxicol* 34(12). <https://doi.org/10.1002/jbt.22597>
- Li H, Kang T, Qi B, Kong L, Jiao Y, Cao Y, Zhang J, Yang J (2016) Neuroprotective effects of ginseng protein on PI3K/Akt signaling pathway in the hippocampus of D-galactose/AIC13 inducing rats model of Alzheimer's disease. *J Ethnopharmacol* 179:162–9. <https://doi.org/10.1016/j.jep.2015.12.020>
- Li H, Song J, Zhang J, Wang T, Yan Y, Tao Z, Li S, Zhang H et al (2017) Ginseng protein reverses amyloid beta peptide and H2O2 cytotoxicity in neurons, and ameliorates cognitive impairment in AD rats induced by a combination of d-galactose and AIC13. *Phytother Res* 31(2):284–295. <https://doi.org/10.1002/ptr.5747>
- Li S, Yu L, Shi Q, Liu Y, Zhang Y, Wang S, Lai X (2022) An insight into current advances on pharmacology, pharmacokinetics, toxicity and detoxification of aconitine. *Biomed Pharmacother* 151:113115. <https://doi.org/10.1016/j.biopha.2022.113115>
- Linjawi SA (2015) Evaluation of the protective effect of panax ginseng nanoparticles against nicotine-induced reproductive disorders in male rats. *J Pharm Sci Rev Res* 32(1):38–45
- Liu F, Grundke Iqbal I, Iqbal K, Gong C (2005) Contributions of protein phosphatases PP1, PP2A, PP2B and PP5 to the regulation of tau phosphorylation. *Eur J Neurosci* 22
- Lü JM, Yao Q, Chen C (2009) Ginseng compounds: an update on their molecular mechanisms and medical applications. *Curr Vasc Pharmacol* 7(3):293–302. <https://doi.org/10.2174/157016109788340767>

- Ludolph AC, He F, Spencer PS, Hammerstad J, Sabri M (1991) 3-Nitropropionic acid-exogenous animal neurotoxin and possible human striatal toxin. *Can J Neurol Sci* 18(4):492–8. <https://doi.org/10.1017/s0317167100032212>
- Mitra A, Rahmawati L, Lee HP, Kim SA, Han CK, Hyun SH, Cho JY (2022) Korean Red Ginseng water extract inhibits cadmium-induced lung injury via suppressing MAPK/ERK1/2/AP-1 pathway. *J Ginseng Res* 46(5):690–699. <https://doi.org/10.1016/j.jgr.2022.04.003>
- Mohammadi Zonouz A, Ghasemzadeh Rahbardo M, Hosseinzadeh H (2023) Antidotal and protective effects of mangosteen (*Garcinia mangostana*) against natural and chemical toxicities: a review. *Iran J Basic Med Sci* 26(5):492–503. <https://doi.org/10.22038/ijbms.2023.66900.14674>
- Mohammadzadeh N, Mehri S, Hosseinzadeh H (2017) Berberis vulgaris and its constituent berberine as antidotes and protective agents against natural or chemical toxicities. *Iran J Basic Med Sci* 20(5):538–551. <https://doi.org/10.22038/ijbms.2017.8678>
- Morsy FA, El Din AAG, Farrag ARH, Badawi MA, Shaffie NM (2012) Ochratoxin A toxic effect on rat kidneys and the potential protective effect of ginseng: histopathologic, histochemical, and image analysis morphometric studies. *Maced J Med Sci* 5(1):40–48. <https://doi.org/10.3889/MJMS.1857-5773.2012.0195>
- Nam OH, Kim JH, Kang SW, Chae YK, Jih MK, You HH, Koh JT, Kim Y (2024) Ginsenoside Rb1 alleviates lipopolysaccharide-induced inflammation in human dental pulp cells via the PI3K/Akt, NF- κ B, and MAPK signalling pathways. *Int Endod J* 57(6):759–768. <https://doi.org/10.1111/iej.14058>
- Nam SM, Choi S-H, Cho H-J, Seo JS, Choi M, Nahm S-S, Chang B-J, Nah S-Y (2020) Ginseng gintonin attenuates lead-induced rat cerebellar impairments during gestation and lactation. *Biomolecules* 10(3):385
- Naraki K, Rameshrad M, Hosseinzadeh H (2022) Protective effects and therapeutic applications of ellagic acid against natural and synthetic toxicants: a review article. *Iran J Basic Med Sci* 25(12):1402–1415. <https://doi.org/10.22038/ijbms.2022.64790.14267>
- Nava-Ruiz C, Méndez-Armenta M, Ríos C (2012) Lead neurotoxicity: effects on brain nitric oxide synthase. *J Mol Histol* 43(5):553–63. <https://doi.org/10.1007/s10735-012-9414-2>
- Park HJ, Kim DH, Park SJ, Kim JM, Ryu JH (2012) Ginseng in traditional herbal prescriptions. *J Ginseng Res* 36(3):225–41. <https://doi.org/10.5142/jgr.2012.36.3.225>
- Park SJ, Lee JR, Jo MJ, Park SM, Ku SK, Kim SC (2013) Protective effects of Korean red ginseng extract on cadmium-induced hepatic toxicity in rats. *J Ginseng Res* 37(1):37–44. <https://doi.org/10.5142/jgr.2013.37.37>
- Park SY, Park JH, Kim HS, Lee CY, Lee HJ, Kang KS, Kim CE (2018) Systems-level mechanisms of action of Panax ginseng: a network pharmacological approach. *J Ginseng Res* 42(1):98–106. <https://doi.org/10.1016/j.jgr.2017.09.001>
- Patrick L (2006) Lead toxicity, a review of the literature. Part 1: exposure, evaluation, and treatment. *Altern Med Rev* 11(1):2–22
- Qi L, Zhang J, Wang J, An J, Xue W, Liu Q, Zhang Y (2022) Mechanisms of ginsenosides exert neuroprotective effects on spinal cord injury: a promising traditional Chinese medicine. *Front Neurosci* 16:969056. <https://doi.org/10.3389/fnins.2022.969056>
- Quan Q, Li X, Feng J, Hou J, Li M, Zhang B (2020) Ginsenoside Rg1 reduces β -amyloid levels by inhibiting CDK5-induced PPAR γ phosphorylation in a neuron model of Alzheimer's disease. *Mol Med Rep* 22(4):3277–3288
- Rajabian A, Rameshrad M, Hosseinzadeh H (2019) Therapeutic potential of Panax ginseng and its constituents, ginsenosides and gintonin, in neurological and neurodegenerative disorders: a patent review. *Expert Opin Ther Pat* 29(1):55–72. <https://doi.org/10.1080/13543776.2019.1556258>
- Rajput SA, Shaikat A, Rajput IR, Kamboh AA, Iqbal Z, Saeed M, Akhtar RW, Shah SAH et al (2021) Ginsenoside Rb1 prevents deoxynivalenol-induced immune injury via alleviating oxidative stress and apoptosis in mice. *Ecotoxicol Environ Saf* 220. <https://doi.org/10.1016/j.ecoenv.2021.112333>
- Ramanathan MR, Penzak SR (2017) Pharmacokinetic drug interactions with Panax ginseng. *Eur J Drug Metab Pharmacokinet* 42(4):545–557. <https://doi.org/10.1007/s13318-016-0387-5>
- Ramesh T, Kim SW, Hwang SY, Sohn SH, Yoo SK, Kim SK (2012) Panax ginseng reduces oxidative stress and restores antioxidant capacity in aged rats. *Nutr Res* 32(9):718–726. <https://doi.org/10.1016/j.nutres.2012.08.005>
- Rameshrad M, Naraki K, Memariani Z, Hosseinzadeh H (2024) Protective effects of Panax ginseng as a medical food against chemical toxic agents: molecular and cellular mechanisms. *Naunyn Schmiedeberg's Arch Pharmacol*. <https://doi.org/10.1007/s00210-024-03186-5>
- Rameshrad M, Razavi BM, Hosseinzadeh H (2017) Protective effects of green tea and its main constituents against natural and chemical toxins: a comprehensive review. *Food Chem Toxicol* 100:115–137. <https://doi.org/10.1016/j.fct.2016.11.035>
- Ramya EM, Kumar GP, Chandrasekhar Y, Anilakumar KR (2022) Adaptogenic potential of ginsenosides against domoic acid-induced toxicity by regulating neuronal stress and kinase receptors: ex vivo and in silico studies. *J Food Biochem* 46(5):e14089. <https://doi.org/10.1111/jfbc.14089>
- Ratan ZA, Youn SH, Kwak YS, Han CK, Haidere MF, Kim JK, Min H, Jung YJ et al (2021) Adaptogenic effects of Panax ginseng on modulation of immune functions. *J Ginseng Res* 45(1):32–40. <https://doi.org/10.1016/j.jgr.2020.09.004>
- Razavi BM, Hosseinzadeh H (2015) Saffron as an antidote or a protective agent against natural or chemical toxicities. *DARU* 23(1):31. <https://doi.org/10.1186/s40199-015-0112-y>
- Royer AST (2023) Copper toxicity. StatPearls [Internet]. StatPearls Publishing, Treasure Island (FL)
- Rushing BR, Selim MI (2019) Aflatoxin B1: A review on metabolism, toxicity, occurrence in food, occupational exposure, and detoxification methods. *Food Chem Toxicol* 124:81–100. <https://doi.org/10.1016/j.fct.2018.11.047>
- Santana-Martínez RA, León-Contreras JC, Barrera-Oviedo D, Pedraza-Chaverri J, Hernández-Pando R, Maldonado PD (2018) Sustained activation of JNK induced by quinolinic acid alters the BDNF/TrkB axis in the rat striatum. *Neuroscience* 383:22–32. <https://doi.org/10.1016/j.neuroscience.2018.04.034>
- Shojaeepour S, Sharififar F, Haghpanah T, Iranpour M, Imani M, Dabiri S (2021) Panax ginseng ameliorate toxic effects of cadmium on germ cell apoptosis, sperm quality, and oxidative stress in male Wistar rats. *Toxin Rev*. <https://doi.org/10.1080/15569543.2021.1884095>
- Shukla R, Kumar M (2009) Role of Panax ginseng as an antioxidant after cadmium-induced hepatic injuries. *Food Chem Toxicol* 47(4):769–773. <https://doi.org/10.1016/j.fct.2009.01.002>
- Simental-Mendía LE, Gamboa-Gómez CI, Guerrero-Romero F, Simental-Mendía M, Sánchez-García A, Rodríguez-Ramírez M (2021) Beneficial effects of plant-derived natural products on non-alcoholic fatty liver disease. *Adv Exp Med Biol* 1308:257–272. https://doi.org/10.1007/978-3-030-64872-5_18
- Singh V, Singh N, Verma M, Kamal R, Tiwari R, Sanjay Chivate M, Rai S N, Kumar A et al (2022) Hexavalent-chromium-induced oxidative stress and the protective role of antioxidants against cellular toxicity. *Antioxidants (Basel)* 11(12). <https://doi.org/10.3390/antiox11122375>

- Siqueira LM, Use COS, Prevention, Ryan SA, Gonzalez PK, Patrick SW, Quigley J, Walker LR (2017) Nicotine and tobacco as substances of abuse in children and adolescents. *Pediatrics* 139(1). <https://doi.org/10.1542/peds.2016-3436>
- Song L, Wang L, Li X, Xiao L (2023) Ginsenoside Rg1 alleviates lipopolysaccharide-induced fibrosis of endometrial epithelial cells in dairy cows by inhibiting reactive oxygen species-activated NLRP3. *Animals (Basel)* 13(23). <https://doi.org/10.3390/ani13233723>
- Song M, Jia F, Cao Z, Zhang H, Liu M, Gao L (2020) Ginsenoside Rg3 attenuates aluminum-induced osteoporosis through regulation of oxidative stress and bone metabolism in rats. *Biol Trace Elem Res* 198(2):557–566. <https://doi.org/10.1007/s12011-020-02089-9>
- Stefanis L (2012) α -Synuclein in Parkinson's disease. *Cold Spring Harb Perspect Med* 2(2):a009399. <https://doi.org/10.1101/cshperspect.a009399>
- Sun M, Ji Y, Li Z, Chen R, Zhou S, Liu C, Du M (2020) Ginsenoside Rb3 inhibits pro-inflammatory cytokines via MAPK/AKT/NF- κ B Pathways and attenuates rat alveolar bone resorption in response to *Porphyromonas gingivalis* LPS. *Molecules* 25(20):4815
- Tabeshpour J, Hosseinzadeh H, Hashemzaei M, Karimi G (2020) A review of the hepatoprotective effects of hesperidin, a flavanone glycoside in citrus fruits, against natural and chemical toxicities. *DARU* 28(1):305–317. <https://doi.org/10.1007/s40199-020-00344-x>
- Tavakkoli A, Ahmadi A, Razavi BM, Hosseinzadeh H (2017) Black seed (*Nigella Sativa*) and its constituent thymoquinone as an antidote or a protective agent against natural or chemical toxicities. *Iran J Pharm Res* 16(Suppl):2–23
- Tian C, Kim YH, Kim YC, Park KT, Kim SW, Kim YJ, Lim HJ, Choung YH (2013) Korean red ginseng ameliorates acute 3-nitropropionic acid-induced cochlear damage in mice. *Neuro-Toxicology* 34(1):42–50. <https://doi.org/10.1016/j.neuro.2012.10.008>
- Wang B, Feng G, Tang C, Wang L, Cheng H, Zhang Y, Ma J, Shi M et al (2013) Ginsenoside Rd maintains adult neural stem cell proliferation during lead-impaired neurogenesis. *Neurol Sci* 34(7):1181–1188
- Wang W, Zhang Y, Li H, Zhao Y, Cai E, Zhu H, Li P, Liu J (2018) Protective effects of sesquiterpenoids from the root of *Panax ginseng* on fulminant liver injury induced by lipopolysaccharide/d-galactosamine. *J Agric Food Chem* 66(29):7758–7763
- Wang X, Wang R, Qiao Y, Li Y (2022) Progress on the efficacy and mechanism of action of panax ginseng monomer saponins treat toxicity. *Front Pharmacol* 13:1022266. <https://doi.org/10.3389/fphar.2022.1022266>
- Wang Y, Wei H, Song Z, Jiang L, Zhang M, Lu X, Li W, Zhao Y et al (2024) Inhalation of panaxadiol alleviates lung inflammation via inhibiting TNFA/TNFAR and IL7/IL7R signaling between macrophages and epithelial cells. *J Ginseng Res* 48(1):77–88. <https://doi.org/10.1016/j.jgr.2023.09.002>
- Xin Q, Ji B, Cheng B, Wang C, Liu H, Chen X, Chen J, Bai B (2014) Endoplasmic reticulum stress in cerebral ischemia. *Neurochem Int* 68:18–27. <https://doi.org/10.1016/j.neuint.2014.02.001>
- Xu Y, Yang L, Liang K, An R, Wang X, Zhang H (2020) Pharmacokinetic effects of ginsenoside Rg1 on aconitine, benzoyleconine and aconine by UHPLC-MS/MS. *Biomed Chromatogr* 34(4):e4793. <https://doi.org/10.1002/bmc.4793>
- Yan X, Chen X, Fu C, Jing C, Zhao D, Sun L (2022) Ginseng oligosaccharides protect neurons from glutamate-induced oxidative damage through the Nrf2/HO-1 signaling pathway. *Food Funct* 13(16):8605–8615. <https://doi.org/10.1039/d2fo01432g>
- Yang X, Chu SF, Wang ZZ, Li FF, Yuan YH, Chen NH (2021) Ginsenoside Rg1 exerts neuroprotective effects in 3-nitropropionic acid-induced mouse model of Huntington's disease via suppressing MAPKs and NF- κ B pathways in the striatum. *Acta Pharmacol Sin* 42(9):1409–1421. <https://doi.org/10.1038/s41401-020-00558-4>
- Yao Y, Long M (2020) The biological detoxification of deoxynivalenol: a review. *Food Chem Toxicol* 145:111649. <https://doi.org/10.1016/j.fct.2020.111649>
- Yi YS (2021) New mechanisms of ginseng saponin-mediated anti-inflammatory action via targeting canonical inflammasome signaling pathways. *J Ethnopharmacol* 278:114292. <https://doi.org/10.1016/j.jep.2021.114292>
- Yildiz M, Boyacioglu M, Avcioglu M, Elmas S (2023) Changes induced by copper toxicity in the rat liver and the effects of panax ginseng on these changes. *Biol Bull* 50(4):S694–S707. <https://doi.org/10.1134/S1062359023604731>
- Yoon SJ, Kim SK, Lee NY, Choi YR, Kim HS, Gupta H, Youn GS, Sung H et al (2021) Effect of Korean Red ginseng on metabolic syndrome. *J Ginseng Res* 45(3):380–389. <https://doi.org/10.1016/j.jgr.2020.11.002>
- You Q, Ma ZC, Wang YG, Hu DH, Liang QD, Xiao CR, Tan HL, Tang XL et al (2015) *Panax ginseng* inhibits intestinal absorption of toxic *Aconitum carmichaeli* alkaloids in vitro. *Chin J Integr Med*. <https://doi.org/10.1007/s11655-015-2282-3>
- Yu C, Xiao J-H (2021) The Keap1-Nrf2 System: a mediator between oxidative stress and aging. *Oxid Med Cell Longev* 2021:6635460. <https://doi.org/10.1155/2021/6635460>
- Yu T, Zhang YX, Liu XJ, Chen DQ, Wang DD, Zhu GQ, Gao Q (2022) Investigation of the pharmacological effect and mechanism of mountain-cultivated ginseng and garden ginseng in cardiovascular diseases based on network pharmacology and zebrafish experiments. *Front Pharmacol* 13:920979. <https://doi.org/10.3389/fphar.2022.920979>
- Yuen Hw BW (2023) Iron toxicity. StatPearls[Internet]. tatPearls Publishing, Treasure Island (FL)
- Zanuso BDO, Santos ARDOD, Miola VFB, Campos LMG, Spilla CSG, Barbalho SM (2022) Panax ginseng and aging related disorders: a systematic review. *Exp Gerontol* 161:111731
- Zhang C, Hou H, Shen C, Ran Q, Cheng F, Yao Z, Zhang R, Peng C (2024) Protective effect of ginsenoside Rb1 against aconitine cardiotoxicity studied by myocardial injury, action potential, and calcium signaling. *Toxicol* 242:107693. <https://doi.org/10.1016/j.toxicol.2024.107693>
- Zhang Y, Pi Z, Song F, Liu Z (2016) Ginsenosides attenuate d-galactose- and AlCl₃-induced spatial memory impairment by restoring the dysfunction of the neurotransmitter systems in the rat model of Alzheimer's disease. *J Ethnopharmacol* 194:188–195. <https://doi.org/10.1016/j.jep.2016.09.007>
- Zhao HH, Di J, Liu WS, Liu HL, Lai H, Lü YL (2013) Involvement of GSK3 and PP2A in ginsenoside Rb1's attenuation of aluminum-induced tau hyperphosphorylation. *Behav Brain Res* 241:228–34. <https://doi.org/10.1016/j.bbr.2012.11.037>
- Zhao L, Zhang Y, Li Y, Li C, Shi K, Zhang K, Liu N (2022) Therapeutic effects of ginseng and ginsenosides on colorectal cancer. *Food Funct* 13(12):6450–6466. <https://doi.org/10.1039/D2FO00899H>
- Zhong S-J, Wang L, Wu H-T, Lan R, Qin X-Y (2019) *Coelogyllum viride* var. *bracteatum* extract improves learning and memory of chemically-induced aging mice through upregulating neurotrophins BDNF and FGF2 and sequestering neuroinflammation. *J Funct Foods* 57:40–47. <https://doi.org/10.1016/j.jff.2019.03.045>
- Zhong SJ, Wang L, Gu RZ, Zhang WH, Lan R, Qin XY (2020) Ginsenoside Rg1 ameliorates the cognitive deficits in D-galactose and AlCl₃-induced aging mice by restoring FGF2-Akt and

- BDNF-TrkB signaling axis to inhibit apoptosis. *Int J Med Sci* 17(8):1048–1055. <https://doi.org/10.7150/ijms.43979>
- Zhou H, Liu Y, Su Y, Ji P, Kong L, Sun R, Zhang D, Xu H et al (2024) Ginsenoside Rg1 attenuates lipopolysaccharide-induced chronic liver damage by activating Nrf2 signaling and inhibiting inflammasomes in hepatic cells. *J Ethnopharmacol* 324. <https://doi.org/10.1016/j.jep.2024.117794>
- Zhu Y, Hu C, Zheng P, Miao L, Yan X, Li H, Wang Z, Gao B et al (2016) Ginsenoside Rb1 alleviates aluminum chloride-induced rat osteoblasts dysfunction. *Toxicology* 368–369:183–188. <https://doi.org/10.1016/j.tox.2016.07.014>

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